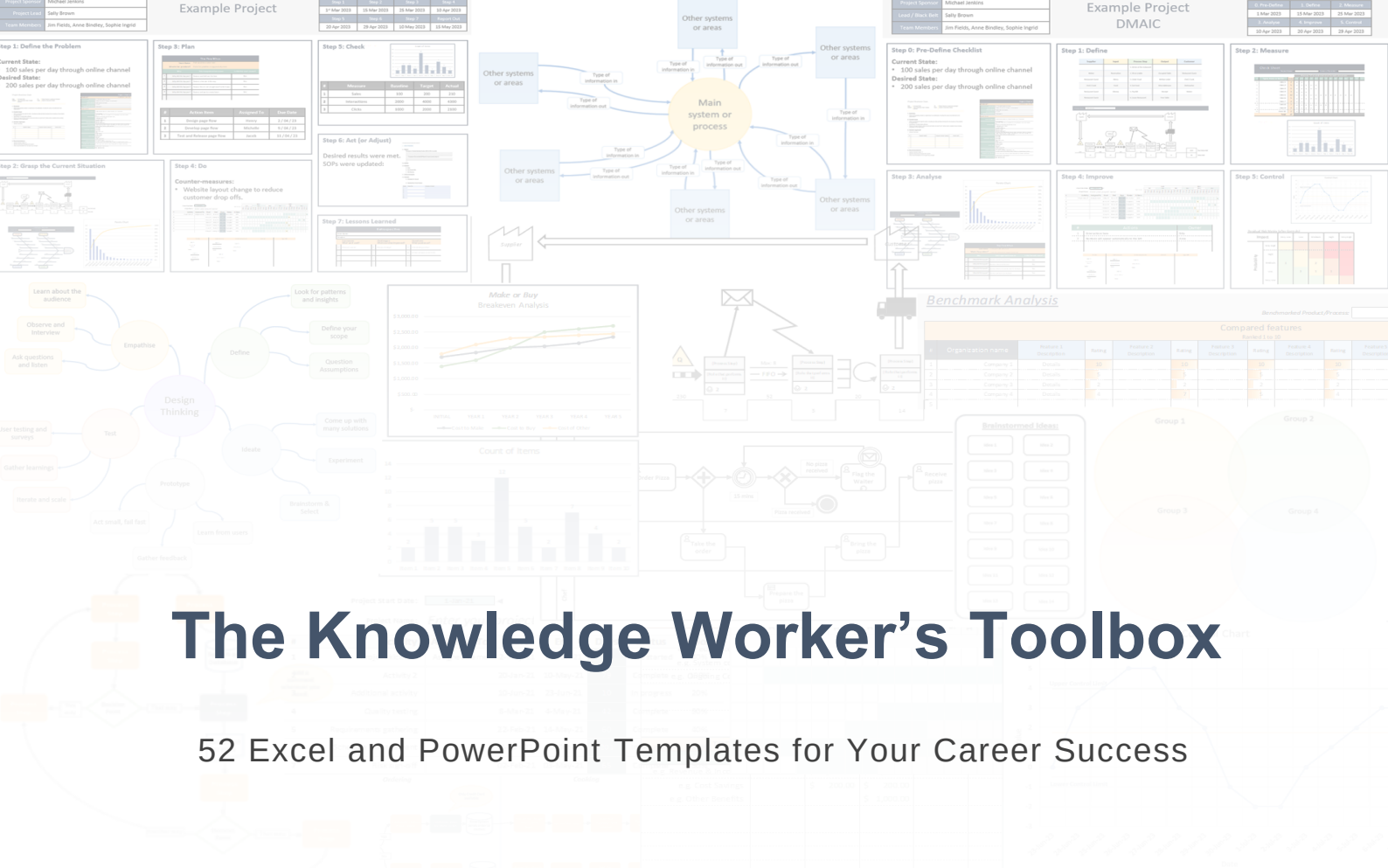




The Knowledge Worker's Toolbox

52 Excel and PowerPoint Templates for Your Career Success



By David McLachlan

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Introduction

Congratulations! You have purchased the absolute finest collection of Excel and PowerPoint templates that will help you get ahead in your job and career. These templates are ready made and ready to use, and will help you in all aspects of knowledge work - from managing a project, leading a team, finding improvements, to visualising a work process or keeping a budget on track. They are also the most-asked-for templates I have experienced in my career as a consultant and project manager.

This collection also represents more than 300 hours of effort to design and create, saving you at least \$4,500 (based on an hourly wage of \$15 x 300 hours) and probably more, if you were to create them all yourself. More than that, they have been used to create millions of dollars' worth of value through projects delivered in companies just like yours.

As an added bonus, this workbook has more detail on each of the 52 templates – from what they are, why we use them, and how to unlock their full advantage to you. They are separated as one per page to ensure it is easy to navigate.

Lastly, I want to wish you the very best for your career. I know these templates will give you the power to create value, lead teams and improve your organisation like never before.

Warm regards,

David McLachlan

Finding and Starting Projects

01. Benchmarking (two templates)

What is it?

Benchmarking is the practice of comparing your business processes or performance metrics to best practices in your industry or other companies. Similar features can also be compared with completely different industries to gather new ideas. Dimensions typically measured are items within quality, delivery time and cost (i.e. how fast is delivery, or how much does this feature cost?). It is particularly useful when forming a business case or finding reasons to initiate a project.

How to use it?

Benchmarking covers a wide range of tools and disciplines. Select your subject and start by defining its process – you might use a Flow Chart, Swim-lane Chart, Value Stream Map or other tool to do this. Make a list of potential partners – remember, they do not have to be in your industry if the process has other similarities. Map out your partner's process, and establish any differences between your two processes, along with the difference in results. Use these differences to brainstorm changes you will implement to close the gap between your two measures.

Some companies may use consultancy firms to gather industry data, as they have access to multiple companies and their details, or databases of industry reports.

When to use it?

Benchmarking is best used during business case or project charter creation, giving weight to the decision to initiate a project and gain initial sign off from your project stakeholders.

Multi-Criteria Benchmarking							
Process Name: Test process			Date Updated: 10-Mar-23				
Rank	3	2	4	1	2	2	
Unique ID	Feature	Company Name	Company Name	Company Name	Company Name	Company Name	Notes
1.0	Feature 1	✓	✗	✗	✓	✗	✓
2.0	Feature 2	✓	✗	✓	✗	✓	✓
3.0	Feature 3	✓	✗	✓	✗	✓	✗
4.0	Feature 4	✗	✓	✓	✗	✗	✗
5.0	Feature 5	✗	✓	✓	✗	✗	✗

Figure 1. Multi-criteria Benchmarking Template

Benchmark Analysis														
Benchmarked Product/Process:														
Compared features														
Ranked 1 to 10														
#	Organization name	Feature 1 Description	Rating	Feature 2 Description	Rating	Feature 3 Description	Rating	Feature 4 Description	Rating	Feature 5 Description	Rating	Feature 6 Description	Rating	Rank
1	Company 1	Details	10		10		10		10		10		10	1
2	Company 2	Details	5		5		5		5		5		5	3
3	Company 3	Details	2		2		2		2		2		2	4
4	Company 4	Details	4		7		5		4		6		8	2
5														
													Total	
													%	
													Rank	

Figure 1.1 Benchmark Analysis Template

02. Cost benefit analysis

What is it?

Cost benefit analysis is a way to measure the benefits you might receive by completing a project, versus the cost of completing it. It is most often measured with money, and as a ratio.

How to use it?

Write down the change you would like to make in your company or business. Note the various benefits you will receive for each, and calculate those benefits over one, two and three years. Benefits might include increased sales (e.g. \$500,000 per year in increased sales), reduced operating costs (e.g. a \$200,000 per year in maintenance savings with increased machine efficiency).

Next write down the cost to complete the project. Depending on how much information you have, you may use the different types of estimating provided in this book, such as bottom-up, analogous, or three point estimating. Lastly, divide the cost into the benefit to give you a ratio (e.g. a \$500,000 benefit divided by a \$200,000 cost equals 2.5, giving us a 1:2.5 cost to benefit ratio. You can also switch it around and show it as a benefit to cost ratio, if you wish.

You may also wish to use “Make or Buy” analysis (shown later in this book) when determining cost – where you check whether it is more efficient to do the change yourself or hire a third party to provide or complete it.

When to use it?

Cost benefit analysis is useful before making a change, when creating a business case or project charter to initiate a project – to check that the project will be worthwhile, and if so to gain the stakeholder sign-offs and proceed.

Cost Benefit Analysis						
Overall Cost to Benefit Ratio: 1 to 2.57						
	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
COSTS						
e.g. Project costs	\$ 1,000.00	\$ 200.00				\$ 1,200.00
e.g. People costs	\$ 200.00	\$ 50.00	\$ 50.00			\$ 300.00
e.g. System costs		\$ 100.00	\$ 100.00			\$ 200.00
e.g. Ongoing Costs		\$ 200.00	\$ 200.00			\$ 400.00
						\$ -
						\$ -
						\$ -
						\$ -
TOTAL	\$ 1,200.00	\$ 550.00	\$ 350.00	\$ -	\$ -	\$ 2,100.00
BENEFITS						
e.g. Revenue & Income		\$ 2,000.00	\$ 2,000.00			\$ 4,000.00
e.g. Cost Savings		\$ 200.00	\$ 200.00			\$ 400.00
e.g. Other Benefits			\$ 1,000.00			\$ 1,000.00
						\$ -
						\$ -
						\$ -
						\$ -
TOTAL BENEFITS	\$ -	\$ 2,200.00	\$ 3,200.00	\$ -	\$ -	\$ 5,400.00
NET BENEFIT OR COST	\$ - 1,200.00	\$ 1,650.00	\$ 2,850.00	\$ -	\$ -	\$ 3,300.00

Figure 02. Cost Benefit Analysis

03. Multi-criteria Decision Chart

What is it?

Multi-criteria decision analysis helps us determine the value of a project in more than just monetary terms. It is a matrix that shows us the different options we have available to us at the top, and the criteria we are ranking against down one side.

How to use it?

Start by brainstorming and writing down the various options you have for your project or for the decision you wish to make, and list them in a heading from left to right. Then brainstorm and write the different criteria you wish to rank them by from top to bottom (for example: initial or ongoing costs, quality or availability of particular features, speed to market). Now you have a table or matrix. Write your rank from 1 to 5 for each option by the criteria in the box that matches, then tally up the total for all the criteria. The ones with the most points (or the highest average) will be your best choices to move forward with.

When to use it?

A multi-criteria chart is useful any time you need to make a decision where multiple criteria drives the outcome. In a project, this is most useful during project initiation when creating a business case or project charter outlining the reasons why your project should begin.

Multi Criteria Decision Chart						
Decision Objective: [Enter the decision to be made here]						
Rate each option from 5 (great) to 1 (poor) depending on their <u>value</u> to you, and use the total at the bottom to assist your decision.						
Criteria	Option 1	Option 2	Option 3	Option 4	Option 5	Option 6
Criteria 1	4	3	2			
Criteria 2	1	5	1			
Criteria 3	5	1	2			
Criteria 4	3	3	4			
Average	3.25	3.00	2.25			

Figure 03. Multi-Criteria Decision Chart

04. Make or Buy Analysis

What is it?

Make or buy analysis is a comparison of two or more options, where we look at the initial and ongoing cost of making something ourselves, versus the initial and ongoing cost of buying something from a Third Party or vendor. It is extremely useful when doing a business case, or determining how to go about acquiring something for your project.

How to use it?

To use “make or buy analysis” we will delve into the “Payback period” as well. Take the initial cost of making something (i.e. a website or a new warehouse), and add the cost of using or maintaining it over time to get your total cost. For example, if we *make* a warehouse it might cost us \$400,000 up front, and \$10,000 a month in ongoing rates and council fees. But if we *buy* warehousing space from a third party it might cost us \$0 up front (we don't have to build it) and \$100,000 a month ongoing. In five months we will have spent \$450,000 by making a warehouse (\$400,000 + \$50,000), or \$500,000 by buying warehouse space (\$100,000 x 5 months). So if we're going to keep it for more than five months it is worth us “making” it instead of “buying” it, and our payback period is five months.

When to use it?

Using Make or Buy analysis is extremely useful when creating a business case to feed into your Project Charter. You can see whether your project should be about making something, or buying something before you even begin. It is also useful when doing your Cost Management Plan and outlining the processes (i.e. Make or Buy) you will use to manage cost in your project.

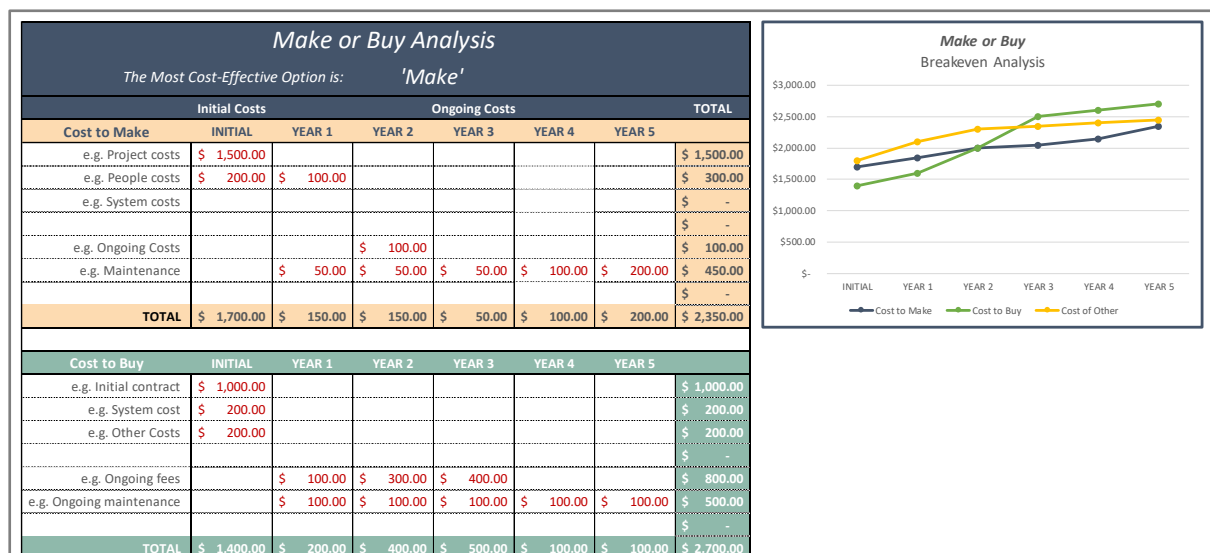


Figure 04. Make or Buy Analysis

05. Business Case

What is it?

A business case assesses a current issue or opportunity, the gap between the current situation and the desired situation, and the proposed solutions to bridge that gap. It is intended as a short document for executive decision makers to approve the recommended investment or course of action – in other words for you to proceed to the next stage of your project.

The next stage is most often gaining project approval through a Project Charter (unless you are working in a PRINCE2 project, where the Business Case acts like the Project Charter).

How to use it?

To write a business case, you will need expertise in the area you are wanting to change or improve. If you don't have it directly, you will need subject matter experts from that area. Keep it to one page, if you can.

First create a short, high level summary of the issue and the opportunity.

Next, define the current issue situation, including any data, facts (such as a feasibility study), known root causes, and stakeholders affected. Note the gap from where we currently are, to where we want to be and advise what is needed to fill that gap.

Next, write the proposed solution ideas, including known risks and whether they are Required, Desired, or Optional.

Lastly, write your recommendations for action to the executive decision makers. This will often mean asking them to support a particular solution, and turn this into a project.

Send an email, or set a meeting with the decision maker (i.e. the executive or manager of that part of the organisation) to walk them through the business case, asking for their action on the recommendations to proceed.

When to use it?

Use a business case before initiating a project, when analysis is needed on an issue that results in a change needing to be made. That change is the solution you will ask for approval to proceed with.

Project Business Case			
Date:	10 August 20xx	To:	Project Sponsor or approving manager
Subject:	Title of Business Case / change required	From:	Analyst or document creator/s
1. Summary – One or two lines: What is the business problem or opportunity to be addressed, including the value to be delivered to the organization?			
2. Current Issue – What is prompting the need for action, including any data and facts showing the root cause(s) of the problem (or opportunity)? – Where is the gap between where we are and where we want to be? Advise what is needed versus existing capabilities of the organization – Identify the stakeholders affected			
3. Solution Approach – Solution Summary			
#	Solution Option	Required, Desired, Optional	Known risks
4. Recommendations – Approve Solution Option 'x' – Approve resources to proceed with solution			

Figure 05. Business Case

06. Business Model Canvas

What is it?

The Business Model Canvas is a single page business plan to help you determine your strengths and weaknesses in a new business, product or feature, value proposition or area. It addresses the most common and highest impact questions you need to answer to ensure your business' success.

How to use it?

Set a meeting with your key business leaders and senior users, to ensure you can brainstorm and capture the information you need. Go through each section of the Business Model Canvas in turn. You can use brainstorming techniques such as the Nominal Group Technique, Affinity grouping, voting, or others you choose. The questions to go through are:

1. Who are the customers we are creating value for?
2. What is the value we are delivering to our customers?
3. What are our revenue streams and revenue models?
4. How do we deliver the value to our customers?
5. What relationship does each customer segment expect from us?
6. What are the key activities we perform each day to create the value?
7. What are the people, resources, systems or money we need to create the value every day?
8. What key partners do we need to deliver the value?
9. What are the costs of creating and delivering value?

When to use it?

Use the business model canvas when you are starting a new business venture or new product. You can also use it when switching products or rearranging parts of an existing organisation.

Business Model Canvas				
Business Name:		Prepared By:		
Key Partners - Which key resource do we acquire from partners? - Which key activities do our partners perform? - Who are our most important partners?	Key Activities What are the activities we perform every day to create and deliver value to our customer?	Value Propositions - What is the value we deliver to our customer? - What customer problems are we solving? - What is the customer need we are addressing? - What is our promise to our customer? - What are the products and services we create for our customer?	Customer Relationships What relationship does each customer segment expect us to maintain?	Customer Segments - For whom are we creating value? - What are the customer segments or groups who pay for or receive our product or service?
	Key Resources What are the resources you need to create & deliver our value proposition?		Channels How does your value proposition reach your customer? Where can your customer buy or use your products or services?	
Cost Structure What are the costs of creating and delivering value to the customer?		Revenue Streams - How do customers pay or reward us for the value we bring them? - What are the different revenue models?		

Figure 06. Business Model Canvas

07. Lean Canvas

What is it?

The Lean canvas is a 1-page business plan template, created by Ash Maurya and adapted from the business model canvas. It is geared towards Lean Start-ups that are creating minimum viable products to release to market, gather feedback on, and pivot in new directions quickly.

How to use it?

Set a meeting with your key business leaders and the team in charge of the area you'll be changing or creating. Go through the sections on the Lean Canvas to brainstorm ideas for your business plan.

1. List your target customers and users.
2. List the top 1 – 3 problems your customer faces.
3. Outline multiple possible solutions for each problem and prioritise the most important.
4. Create a clear Value Proposition – why you are different and worth paying attention to.
5. High Level Concept – list your analogy – What is it a combination of?
6. Who will be the early adopters?
7. What channels will you deliver the value through?
8. What is the revenue model for your product?
9. What are the costs involved in creating and delivering your product?

When to use it?

Use the Lean canvas as an alternative to the business model canvas, when you are shifting strategies, creating a new product, or starting a new business.

Lean Canvas Business Plan				
Business Name:		Prepared By:		
Problem ▶ List your top 1 - 3 problems.	Solution ▶ Outline a possible solution for each problem.	Unique Value Proposition ▶ Single, clear, compelling message that states why you are different and worth paying attention to.	Unfair Advantage ▶ Something that cannot be easily bought or copied.	Customer Segments ▶ List your customers and users.
Existing Alternatives ▶ List how these problems are solved today.	Key Metrics ▶ List the key numbers that tell you how your business is doing.	High Level Concept ▶ List your X for Y analogy i.e. YouTube = Flickr for videos.	Channels ▶ List your path to customers (inbound or outbound)	Early Adopters ▶ List the characteristics of your ideal customers.
Cost Structure ▶ List your fixed and variable costs.		Revenue Streams ▶ List your sources of revenue.		

Figure 07. Lean Canvas

08. Project Charter

What is it?

The Project Charter is one of the first documents you will need for your project, after the business case. It is often either created by the project sponsor (or their area), or signed off by their area, and is used to gain agreement on the project and buy-in from the necessary stakeholders or business owners.

Because it is done at the very beginning it is a high-level document that gives a broad idea of key project items such as business requirements or needs, schedule or milestones, risks, assumptions, stakeholders, and the business case (including return on investment).

Other items you can include are the project purpose, measurable objectives or success criteria, key deliverables, pre-approved or necessary financing, resources required, assigned project manager with responsibility and authority level, and the name and authority of the sponsor (or people authorising the project).

How to use it?

You will use the project charter differently depending on your role in a project. Sometimes it is created by the project sponsor or within their team and it needs approval by a Project Management Office or executive team. Other times it is created by the project manager or someone external and the Project Sponsor will need to agree and sign off on the proposal. Each organisation will be different.

When creating a Project Charter find and add each of the items above (stakeholders, requirements, schedule, risks, resources, cost etc). The main input will be the business case, such as a Cost Benefit analysis. Ensure you have expertise from the sponsor's area, as this will help smooth their decision. Place all the items in to the Project Charter, then set a meeting with the Project Sponsor to sign off.

Walk them through the proposal, including the business need, costs and benefits, resources required. Ask for their agreement, and advise the next steps to proceed. Gain their sign off in writing (often by email), then you can go ahead with your project.

When to use it?

Use a Project Charter at the very beginning of your Project. It is the gateway into a sponsor's business area – ensuring they understand the reasons, the benefits, and then agree to the use of their resources or funding. A good Project Charter will set the stage for a well-run project with the proper authority and resources in the future.

Project Charter	
GENERAL PROJECT INFORMATION	
PROJECT NAME	e.g. Online Self Serve Payment System
PROJECT SPONSOR	The executive paying for the project, or giving resources.
PROJECT MANAGER	The person leading the project.
EXPECTED START DATE	01 June 23
EXPECTED COMPLETION DATE	30 September 23
PROJECT DETAILS	
PROJECT DESCRIPTION	High-level project description, purpose and reason "why" for the project.
KEY REQUIREMENTS	What is required? Also written as "Features", a high-level list of what is required as an output of this project. e.g. The team will deliver these usable features in order: 1. Webpage – working design 2. Online Payments 3. Back-end Integration
ESTIMATED MILESTONES	Feature 1 – June 30 Feature 2 – July 30 Feature 3 – September 10
EXPECTED BENEFITS	Enhanced customer experience providing access to an online channel for our customers, thereby personalising their channel of choice. Reduction in wait times for customers. Staff savings of around 9,960 hours of effort each year.

Figure 08. Project Charter

09. Team Charter

What is it?

The Team Charter is an agreement within your team on how you will all interact and work together. It answers the questions: “Why are we doing this project? Who benefits and how? What does ‘Done’ mean for this project? and How are we going to work together?” It is most commonly used within Agile teams or on Agile projects.

The Team Charter typically includes things like the team values, communication guidelines, decision making criteria or process, conflict resolution process, meeting guidelines or other team agreements.

How to use it?

Set a meeting with your team before your project begins or as soon as it begins, with the agenda to discuss the Team Charter. Go through and come to a majority agreement on each of these points, and any others you would like to add:

What are the team values?

How will the team communicate?

How are decisions made?

How is conflict resolved?

How often do we meet, and what are meeting rules? For example are daily stand-ups mandatory to attend, only one person speaks at a time, the team takes turns to facilitate, agendas must be clear in meeting invites.

When to use it?

A Team Charter is most commonly used at the beginning of your project, then referred to and refined over the life of your project. By sitting with the team and deciding on your way of work early, it reduces future conflicts. And by refining the team charter after receiving feedback (often through a Retrospective), you ensure it remains relevant to the team.

Project Charter	
GENERAL PROJECT INFORMATION	
PROJECT NAME	e.g. Online Self-Serve Payment System
PROJECT SPONSOR	The executive paying for the project, or giving resources.
PROJECT MANAGER	The person leading the project.
EXPECTED START DATE	01 June 23
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PROJECT DETAILS	
PROJECT DESCRIPTION	High-level project description, purpose and reason “why” for the project.
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ESTIMATED MILESTONES	Feature 1 – June 30 Feature 2 – July 30 Feature 3 – September 10
EXPECTED BENEFITS	Enhanced customer experience providing access to an online channel for our customers, thereby personalising their channel of choice. Reduction in wait times for customers. Staff savings of around 9,960 hours of effort each year.
STAKEHOLDERS (IMPACTED)	Name, Role Name, Role Name, Role Department or area Team
PROJECT TEAM	Project Sponsor: Project Manager: Business Analysts: Developers: Test Lead: Subject Matter Experts:
PROJECT EXIT CRITERIA	Also “Success Criteria” e.g. Working, usable features released on time. What are the conditions to be met in order to close or to cancel the project or phase.
ESTIMATED COST & RESOURCES	e.g. 39k costs including utilisation of internal delivery team and Subject Matter Experts
OVERALL PROJECT RISK	High level, what could go wrong? Risk, Likelihood, Impact Risk, Likelihood, Impact

Figure 09. Team Charter

10. DMAIC

What is it?

DMAIC is a method of problem solving and project delivery popularised by Motorola and Six Sigma. As an iterative approach very similar to PDCA, it stands for:
Define, where we define the problem clearly,
Measure, where we gather data about the problem,
Analyse, where we analyse the root cause of the problem based on that data,
Improve, where we put our recommendations in place,
Control, where we check the data to see if we got the results we wanted, and adjust as necessary.

How to use it?

First note the business problem you would like to work on and solve. Create a document or plan with a space for each section of DMAIC above, then add your detail to each section (for example the clear definition of the problem and the reason behind solving it, the data and measures, the analysis of root cause, and your improvement recommendations). It is important to note that there are many smaller methods within each DMAIC section (i.e. histograms, Ishikawa diagrams), using the project tools you will learn in this book.

When to use it?

Use DMAIC when you are delving into a business problem and the reasons it occurs. For this reason it is useful during business case creation, or when something in your project or business is going off track (such as schedule, cost, or quality).

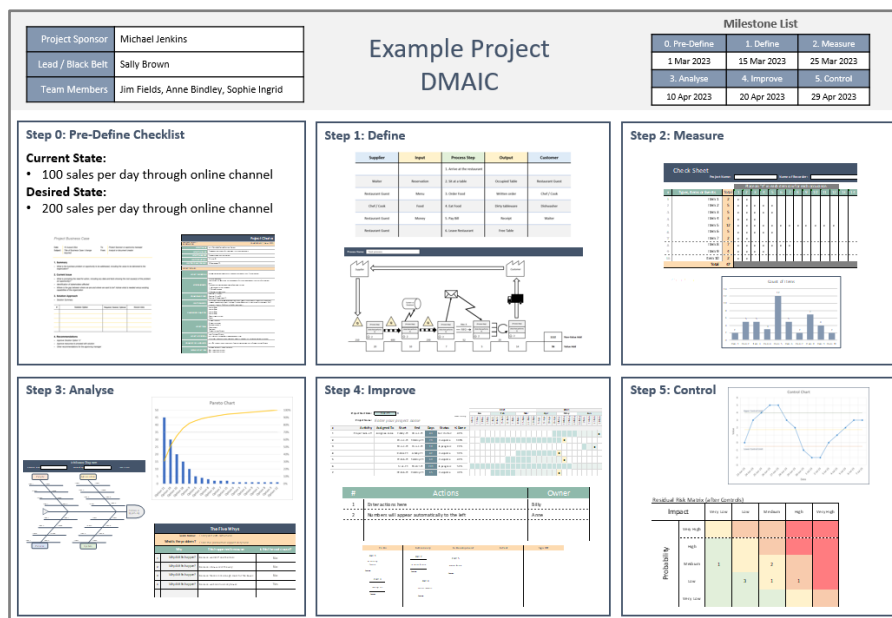


Figure 10. DMAIC on a page

11. Plan Do Check Act

What is it?

Plan Do Check Act, or Plan Do Study Adjust is an iterative method of product development or change. It states that we first observe the current situation and Plan what we are going to do. Then we perform the plan – “doing” it. Then we Check that we got the results that we wanted. Finally we adjust based on the results we got, and redo the plan to begin the cycle again. It is the precursor to many Agile methods such as iterative product development, retrospectives and so on.

How to use it?

First observe the current situation you are wanting to change. Gather data about the problem so the current situation and problem are clear, and use this to write your plan. This plan is similar to a business case or even a project charter, often on a smaller scale, and it describes what we are going to do about the problem. Then perform the plan, check its results and perform a retrospective afterwards where you ask the team how we could improve. Use that feedback to write your new plan and begin again.

When to use it?

PDCA can be used when improving the quality within your project or business – both on the project process itself through retrospectives, and on the project product or service you are using or releasing. For this reason it is useful during Quality Assurance or testing of your product, and in meeting or resource management for your project team.

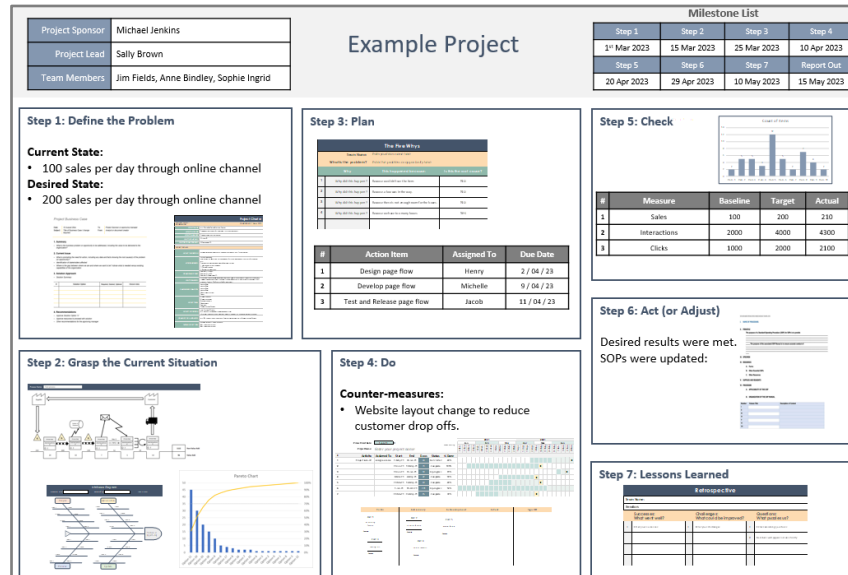


Figure 11. Plan Do Check Act on a page

Gathering Scope and Requirements

12. Six Thinking Hats

What is it?

The Six Thinking Hats is a framework by Edward De Bono for brainstorming ideas, opportunities and risks in your project. It takes a central question or idea (such as a new product) and then looks at it from different points of view – the different coloured “hats”.

How to use it?

Set a meeting with your team or stakeholders who have expertise in the idea you wish to brainstorm about (i.e. for a new product – the business analyst, project sponsor, testers or other key subject matter experts). Write the key idea on a whiteboard, and the hat colours below them. Then either assign the following hats to groups, or go through them all together one by one:

White hat – purely the facts and figures about the idea.

Red hat – our emotions or feelings about the idea, purely subjective without needing to justify them.

Black hat – noting the risks, dangers, and what could go wrong.

Yellow hat – optimistic and constructive proposals for making things happen.

Green hat – for creative ideas, thinking outside the box, what would we do if we could do anything?

Blue hat – to summarise and conclude it all with next steps and a way forward

When to use it?

You can use the Six Thinking Hats during the Planning phase of your project, in the Scope and Risk knowledge areas. It will help you come up with new product ideas for your business, new improvement opportunities, or new features with your customer or project sponsor. You can also use it to gather risks and controls when planning the Risk phase of your project.

Brainstorming with the Six Thinking Hats			
Hat Type	Description	Ideas	Priority
White Hat 	Purely the facts and figures about the idea. What do we know for sure?	Idea 1	4
		Idea 2	3
		Idea 3	2
Red Hat 	Emotions, purely subjective without needing to justify them. What do we feel?	Idea 1	1
		Idea 2	5
		Idea 3	6
Black Hat 	Risks, dangers, and anything that could go wrong.	Idea 1	9
		Idea 2	8
		Idea 3	7
Yellow Hat 	Optimistic, constructive proposals for making things happen.	Idea 1	11
		Idea 2	12
		Idea 3	10
Green Hat 	Creative ideas and thinking outside the box. What would we do if we could do anything?	Idea 1	15
		Idea 2	14
		Idea 3	12
Blue Hat 	Summarise and conclude it all with next steps and a way forward.	Idea 1	17
		Idea 2	16
		Idea 3	18

Figure 12. Brainstorming with the Six Thinking Hats

13. Nominal Group Technique

What is it?

Nominal Group Technique is an anonymous form of brainstorming, to help reduce the impact that others may have in swaying group decisions.

How to use it?

When we have something that needs to be decided, a question is posed to the group. Each person silently writes down their ideas or answers. The facilitator then takes the ideas and writes them on the whiteboard – sometimes grouping them using an “Affinity Diagram” or method. The group then votes for the idea they want, but privately – often by writing it on a piece of paper then handing them in to the facilitator.

When to use it?

You can use Nominal Group Technique any time a decision needs to be made – on project scope, on proposed procurements or third parties, on project risks, on which method to use, or what to deliver next. It is a handy technique that can be used almost anywhere.

Nominal Group Technique - Multi Voting									
Each team member has: 10 points to distribute on ideas.									
Facilitator:									
Date: 21-Mar-23									
Allocated Points									
Brainstormed Ideas List	Team Member 1	Team Member 2	Team Member 3	Team Member 4	Team Member 5	Team Member 6	Team Member 7	TOTAL	RANK
Brainstormed Idea 1	4	1	1					6	4
Brainstormed Idea 2	2	3	3					8	3
Brainstormed Idea 3	1	5	1	4				11	1
Brainstormed Idea 4	1	1	5	3				10	2
Brainstormed Idea 5	1			2				3	5
Brainstormed Idea 6	1			1				2	6
Brainstormed Idea 7				1				1	7

Figure 13. Nominal Group Technique – Multi Voting

14. Affinity Diagram (two templates)

What is it?

Affinity diagrams allow a large number of ideas or items to be classified into similar groups. They are useful for taking an overwhelming number of things and grouping them together in ways that make sense for your project.

How to use it?

Start by gathering the ideas you want to group together – usually by brainstorming with your team. Ensure you can move any written ideas around, for example use sticky-notes in a physical environment, or a virtual whiteboard in a virtual environment. Go through the list with your stakeholders, and for each one check whether it is similar or related to another. If it is, place them next to each other.

You may need to use voting techniques such as the Fist of Five or Nominal Group Technique if the team can't immediately agree, and to finish up you can circle and name your groups as the team sees fit.

When to use it?

You will often use an Affinity Diagram during a brainstorming session with your project team, when it makes sense to group a large amount of ideas together. These times might include gathering the scope and features of your project, where you are able to place the many ideas your stakeholders come up with into similar-featured groups that can then be worked on at the same time. Another time might be during Quality or testing of your product, where you gather any defects found and group them together to show to your project team and sponsor.

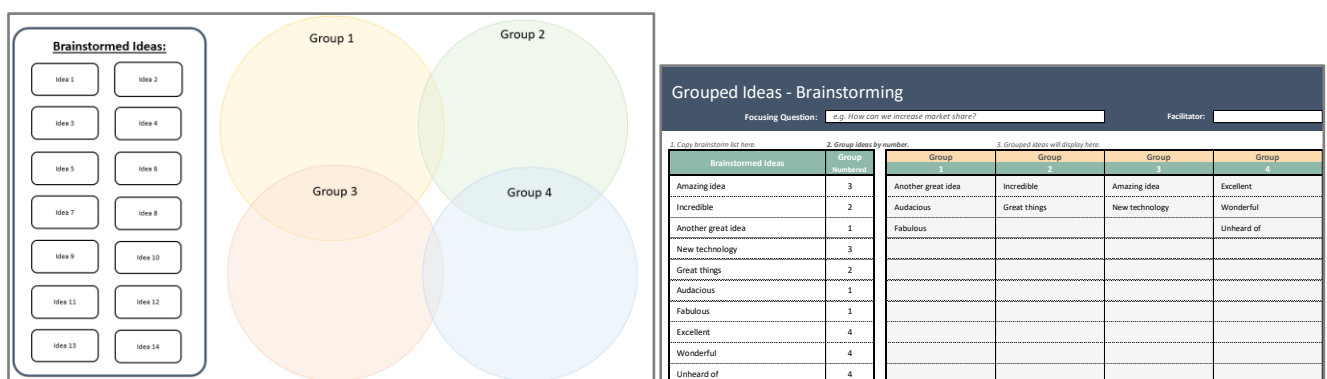


Figure 14. Affinity Diagram and Brainstorming

15. Mind Mapping

What is it?

A mind map is a hierarchical diagram used to show relationships between ideas – usually starting with a single, central concept, where related ideas branch off from. It is useful for showing how things relate to each other, or for brainstorming further concepts down different streams of thought.

How to use it?

Start by drawing a circle in the centre of your page or whiteboard, and write the central theme or idea you wish to brainstorm ideas about. For each idea you come up with, draw another circle, write the idea, and link it to the main circle with a line. Do the same for any extra ideas that stem from those new circles – moving further and further out.

When to use it?

Drawing a mind map is useful when gathering requirements, discovering a process or area, or creating a new product as it allows you to brainstorm additional things around a central idea. It is also useful when visualising the teams you are working with or the way your project team works, in lieu of an Organisational Breakdown Structure chart.

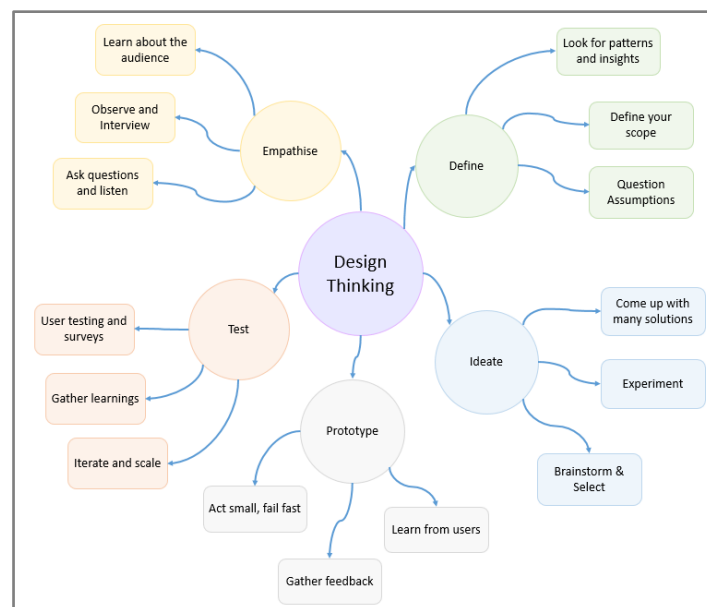


Figure 15. Mind Mapping

16. Context Diagram

What is it?

A Context Diagram can also be called a “Data Flow Diagram”, and that gives us a clue to its use. It shows a core system or process, the systems or things it interacts with, and the type of data that flows between them.

How to use it?

Start by drawing a circle in the centre of your page, and note the product, main system or process you'll be analysing. Then draw boxes around the circle that represent external agents - other systems, where information flows to or from.

Finally, draw arrows to or from your main system, noting the information that flows on that path.

When to use it?

Use the Context Diagram to start the Requirements and Scope gathering on your project. It is useful to show the current state of a system or process, the future state of a system or process, and to get a high level view of where we may need to gather further information from.

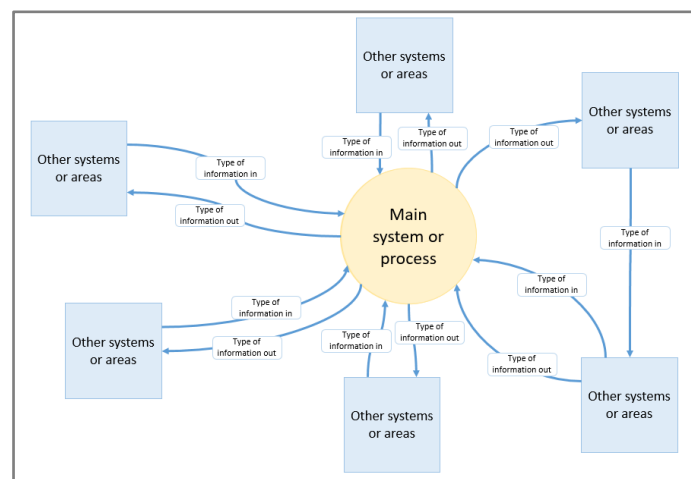


Figure 16. Context Diagram

17. SIPOC

What is it?

A SIPOC is an acronym that stands for “Supplier, Input, Process, Output, Customer”. It is a way of capturing a process and its associated Suppliers (what comes before), Customers (what comes after) and their various inputs.

How to use it?

Start by creating your five boxes, one for each letter, then set a meeting with the person or team that knows the process the best – you will need their expert judgement. When everyone is together, write down the process you wish to analyse in the “Process” section. Then ask the team to brainstorm the:

“Output” of that process – what the output of the process is,

“Customers” of that process – who receives the output of it,

“Suppliers” of that process – who provides input into the process,

“Input” to the process – what inputs are needed to make the process happen.

When to use it?

A SIPOC is an ideal tool to use when you are figuring out the current state and how things currently happen. You can also create a “Future State” SIPOC to show how something will work in the future. Because of this, a SIPOC works during the Scope phase of a project to gather requirements. You might use it in a business case before creating a Project Charter, and you could use it to inform the risks of a project if needed.

Supplier	Input	Process Step	Output	Customer
Supplier Name (Who provides input into the process)	Input Name (What is needed to perform the process)	1. Arrive at the restaurant	Output Name (the process outcome or deliverable)	Customer Name (who benefits or receives the output)
Waiter	Reservation	2. Sit at a table	Occupied Table	Restaurant Guest
Restaurant Guest	Menu	3. Order Food	Written order	Chef / Cook
Chef / Cook	Food	4. Eat Food	Dirty tableware	Dishwasher
Restaurant Guest	Money	5. Pay Bill	Receipt	Waiter
Restaurant Guest		6. Leave Restaurant	Free Table	

Figure 17. SIPOC

18. Value Stream Mapping

What is it?

A Value Stream Map is a visual way of showing how value is created and delivered to a customer through process steps, picture icons and visual representations of “waste” from a Lean point of view. It revolves around the customer – starting with a customer order, showing the process steps to create that value, and showing how it is delivered.

How to use it?

Start by writing the name of the customer you are delivering to in the top right, remembering that a “Customer” can be anyone internal or external who receives the benefit of something you do. Write the trigger or the way the process starts in the top left, and link the customer to that trigger. Then draw boxes down the bottom from left to right for the process steps, ultimately looping it all back together by delivering to the customer in the top right again.

There are also various icons for things like Kanban, Supermarkets, Pull, Push, inventories and queues, transport and more. These are Lean ideas and methods you may be implementing in your work environment, and a full list is provided in the Value Stream Map template.

When to use it?

A Value Stream Map is very useful before your project is initiated, to map the current process and discover the business opportunities you want to solve. Those opportunities will go into your business case and Project Charter. It is also useful during the “Scope” knowledge area of your project, commonly used by the Business Analyst when gathering the current situation, customer requirements and Project Scope.

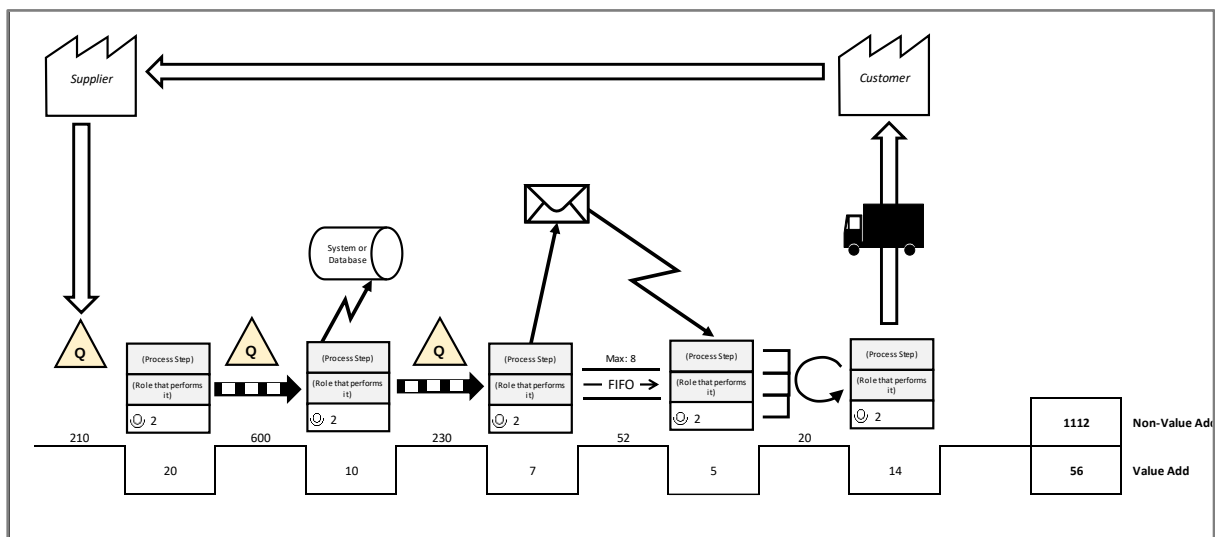


Figure 18. Value Stream Mapping

19. Flow charts (two templates)

What is it?

A flow chart is a way of visualising something that flows from beginning to end. This could be a current or future process you are working with in your project, or it could be the way something currently happens from start to finish.

How to use it?

There are three main icons in a flow chart – a rounded rectangle to start and finish, and square for a process step, and a diamond for a decision point.

Set a meeting with the key knowledge experts in the area you wish to map. Remember – you don't need to be an expert in everything but you will need to find, coordinate and facilitate the experts to help you do this. Start at the beginning with a rounded rectangle and write "Start" inside it. Then write out the square process steps with the process name inside them, and link them with arrows. For places where it could flow two or more ways, draw a diamond with the key question that determines which way it will go, and write the answer on the line that goes to the next process step. Finish your chart with a rounded rectangle again.

When to use it?

A flow chart is useful to show the current and proposed future states of a process, visualising it during the requirements and scope gathering phases of your project. You can also use it when creating a business case, or just after that with a project charter – as you can show your stakeholders the reason why a project is important and needs to be initiated.

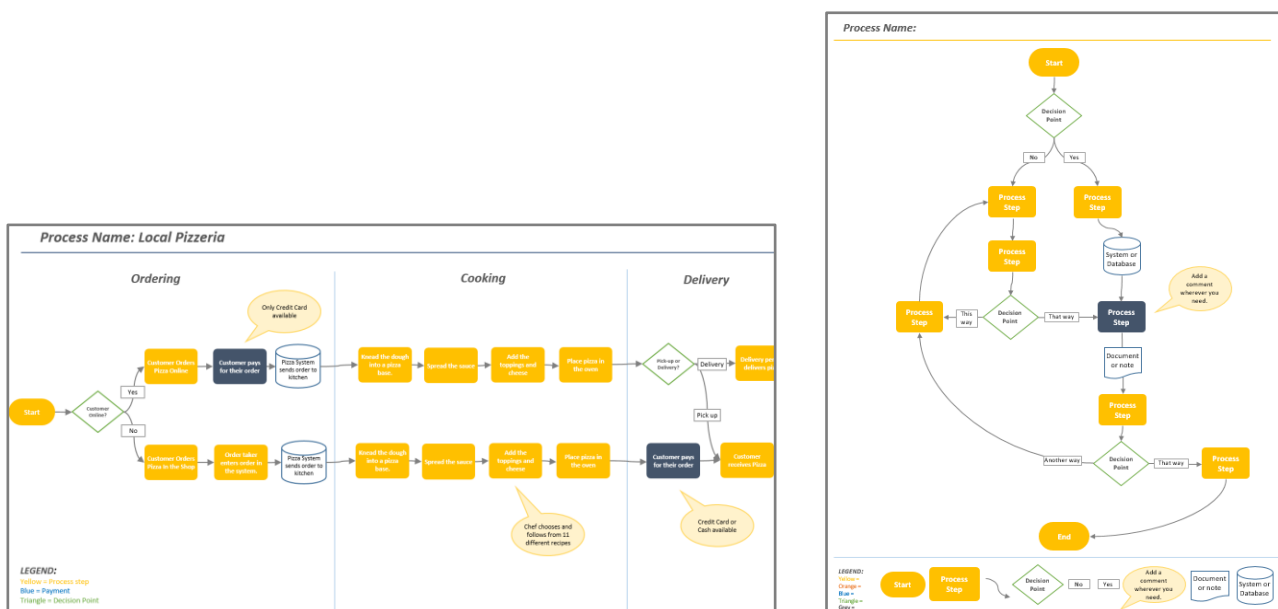


Figure 19. Flow Charts – Horizontal and Vertical

20. Swimlane chart

What is it?

A swimlane chart is process flow chart, and is a way to clearly show process steps and how they flow between different people, areas or departments. It is useful to see the hand-off points in a process, where a process might be being held up or where it could be improved.

How to use it?

Start by noting the process you want to capture (e.g. “accounts receivable”), then write all the people or areas involved in the process, from top to bottom on the left side of your page. Each of these will be their own swim-lane. Then draw a box in the swimlane where the process starts and write the first process step (e.g. “Create invoice” in the “Invoicing Department” lane). Map out the process from left to right, with new boxes for each step, until you reach the end. If you need a “decision point”, use a diamond shape.

When to use it?

Swimlane charts are useful during project initiation and planning. You can gain support for your project by showing your stakeholders the current process and the pain points you might be trying to solve. During project planning it is useful for managing requirements and scope in your project – often by a business analyst, by showing the current or future states.

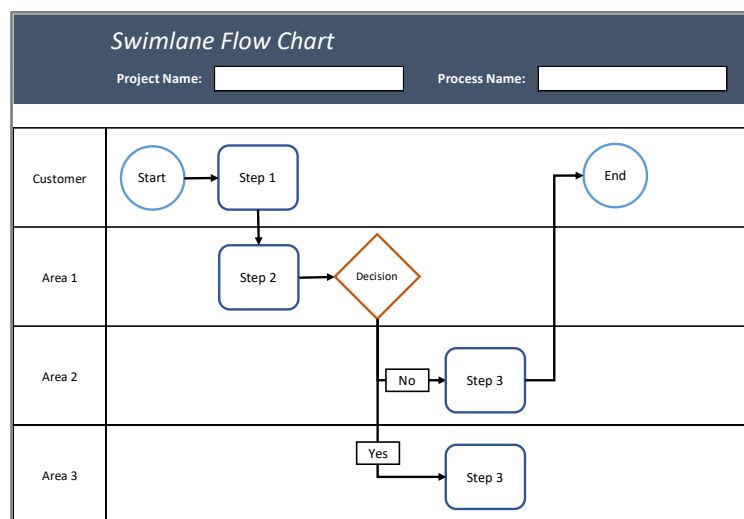


Figure 20. Swim Lane Flow Chart

21. BPMN 1.0 and 2.0

What is it?

BPMN stands for Business Process Model and Notation, and is a more in-depth type of flow chart that incorporates the ideas of a swim-lane chart with basic flow chart symbols and adds more detailed symbols and diagrams for additional process information. In some cases processes will also be linked electronically using software and include detailed process information such as timings, dependencies and owners.

How to use it?

Similar to the other process chart methods, start by noting the process you want to capture. This will be its own “Pool”, where you note the different areas or people involved down the left-hand side. Write the process steps in boxes from left to right in the rows that coincide with the people performing them. Add further detail according to the accepted BPMN icons shown in the template.

When to use it?

BPMN is useful when trying to visualise the way things are currently done, or the way things could be done. Use it when creating your business case or project charter and initiating your project, to show your stakeholders the pain points that need to be solved. You can also use it during the planning and validation stages for your project scope, to show any changes to your stakeholders.

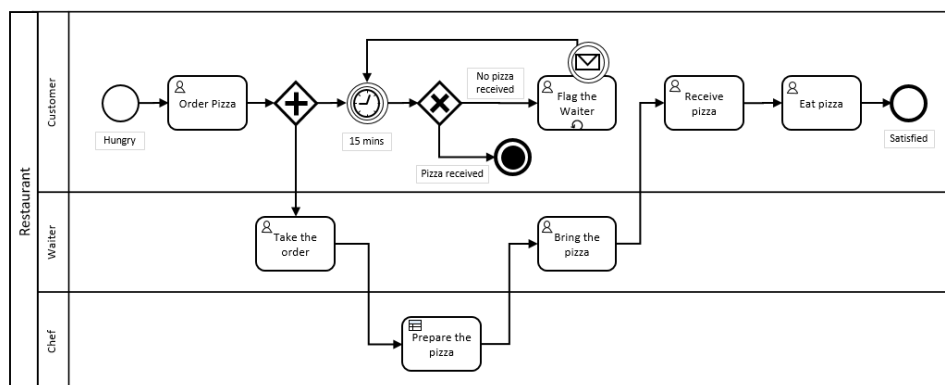


Figure 21. BPMN Process Chart

22. MoSCoW (and Trade-off Sliders)

What is it?

MoSCoW analysis is an acronym, standing for Must-have, Should-have, Could-have, and Won't-have. It is an easy acronym to remember, and is a way to prioritise requirements or scope in your project from most important to least important.

How to use it?

Gather a list that you would like to prioritise – most likely the list of requirements from your customer (internal or external such as your project sponsor or paying customer), or the defined scope that you will deliver as part of your project.

Set a meeting with the key decision makers for these items – most likely the project sponsor, developers or builders, and key experts assigned from that area.

Go through the list one at a time, noting its priority. To simplify, some managers just use Must-have, Should-have, or Nice-to-Have. You can use voting techniques such as the Fist of Five (also in this book) if the group cannot agree.

When to use it?

You can use MoSCoW to prioritise the Requirements and Scope on your project as early as the Project Charter or business case, and throughout the Scope Management phase. It is useful for knowing what to focus on first, and what items will bring the most customer happiness.

MoSCoW				
Project Name:				
Must have Should have Could have Won't have				
Place an X in the appropriate cell - one per row				
Requirements	Must have	Should Have	Could Have	Won't Have
[Enter requirement here]	X			
[Enter requirement here]		X		
[Enter requirement here]			X	
[Enter requirement here]				X
[Enter requirement here]	X			
[Enter requirement here]			X	

Trade-Off Sliders					
Place one "X" in each row					
	Value	Cost	Time	Scope	Quality
Most Fixed	X				
					X
		X			
				X	
			X		
Most Flexible					

Figure 22. MoSCoW and Project Trade-off Sliders

23. Kano Analysis

What is it?

Developed in the 1980s by Professor Noriaki Kano, the Kano Model separates product features into three main categories: Must-Be Quality (Basics), One-dimensional Quality (Satisfiers), and Attractive Quality (Delighters).

Basic features are the absolute basics – if they're not there a customer will complain. Satisfiers are features that will increase the customer satisfaction. Delighters are providing something that customers haven't asked for, but that makes the product even better. A customer will not usually complain if Delighters are not there. Because customer needs evolve over time, Delighters will often become Satisfiers, and Satisfiers will soon become Basics.

How to use it?

Gather the list of high-level features, customer requirements or project scope with your customer or project sponsor. Set a meeting with the key decision makers on your project, such as the project sponsor (or assigned experts from their area) customers, and key developers.

Go through the list and decide whether each is Basic (the customer complains if not there), Satisfier (customer satisfaction increases), or Delighter (Customer raves about it). Remember that Basic features must be met before considering Delighters.

You may need to use a voting technique for this process if people have differing opinions, such as the Fist of Five or Nominal Group Technique.

When to use it?

Kano Analysis is useful during the Initiating and Planning stages of your project, especially for prioritising Customer Requirements or Project Scope. It can be done as part of the business case or Project Charter, or throughout the work on your Project Management Plan.

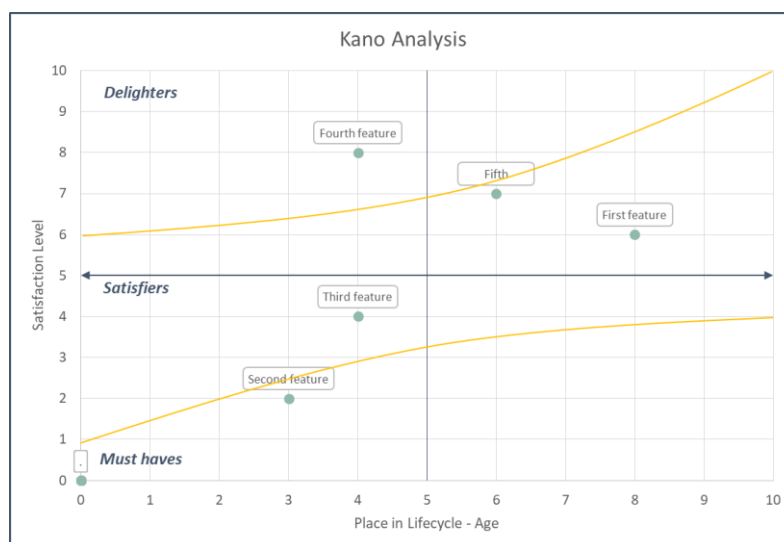


Figure 23. Kano Analysis

24. Work Breakdown Structure (two templates)

What is it?

A work breakdown structure uses “decomposition” to break down high level features or ideas into more manageable deliverables that can be assigned to your project teams to estimate on. Keep in mind it does not result in the activities, just smaller deliverables.

How to use it?

Start with the high-level feature or list of features that you want to deliver. For each feature, write them in a box – you will break them down underneath each box. Set a meeting with the people who will understand the work required – it might be the business owner, the team involved in doing it (i.e. developers, testers, business analysts in an I.T environment, or architects, designers, builders in construction). Break the high-level feature into smaller pieces and write those pieces in new boxes underneath that feature. Break it down again if required until they are actual work packages that a person or a team can work on and deliver.

You can add more detail to each item with a **WBS Dictionary** table, where you include a description of the work, dependencies, schedule milestones, resources required, cost estimates, quality requirements, test criteria and sign offs.

When to use it?

Creating the Work Breakdown Structure is done during project planning in the Scope Management knowledge area of your project. It doesn't have to be done just once either – you might adjust or use an iterative process over time if features or other things change. In a project delivered using Agile you might use “Rolling Wave Planning”, where the high-level feature is broken down into smaller workable packages and estimated on just before an iteration.

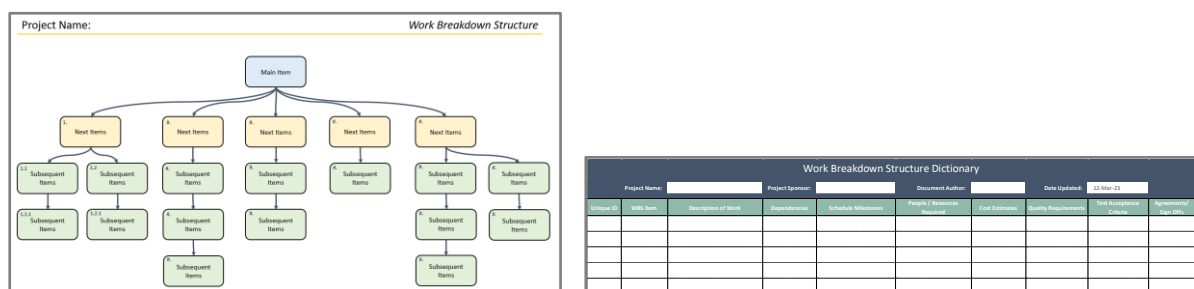


Figure 24. Work Breakdown Structure and Dictionary

25. Requirements Traceability Matrix

What is it?

The requirements traceability matrix is a grid that links product requirements to the project deliverables and scope that satisfy them. It provides a means to track requirements throughout the project life cycle, helping to ensure that those requirements are delivered at the end of the project. It also provides a structure for managing changes to the product scope.

How to use it?

Draw a grid with item I.D.s on the left and the Requirements Description. Then note the Business Need (reason why), WBS Deliverable that meets that requirement, definition of done, and Test Case ID. You may add other columns as needed. Now list the requirements gathered for your project from top to bottom and grouped as necessary, and fill out the remaining columns to suit. You may need the expert judgement of various project team members such as the business analyst, quality tester, project sponsor or business owner.

When to use it?

The requirements traceability matrix is best used when planning the scope of your project, and throughout the execution and monitoring of your project when validating scope and checking the quality of the deliverables.

Requirements Traceability Matrix						
Project Name:			Document Author:			
Req # Unique ID	Requirement Name / Description	Scope Deliverable or Feature	User Story or Work Packages	Assigned To	Test Case	Current Status

Figure 25. Requirements Traceability Matrix

26. Customer Journey Map

What is it?

A customer journey map is a visual representation of the customer journey (also called the buyer journey or user journey), showing the highs and lows of their interaction. It helps you tell the story of your customers' experiences with your product or service across all touchpoints.

How to use it?

A customer journey map will most often start with a Customer Persona – a placeholder profile of the typical customer's age, job, family and personal goals that will help you step into the customer's shoes.

To make your customer journey map you will need a few things. First, note the system or experience you wish to analyse, and the touch points or areas that customers interact with. To gather the right data of their experience you will need to interview customers, hold focus groups, or research Voice of Customer (VoC) comments and data.

Note the areas from left to right, with touchpoints for each area underneath. For each touchpoint or interaction, note the customer's rating out of 10 and their specific comments, then map the highs and lows against each one.

When to use it?

Use a customer journey map when you want to understand where to improve your product, system or service, and find new areas of value you can bring to market.

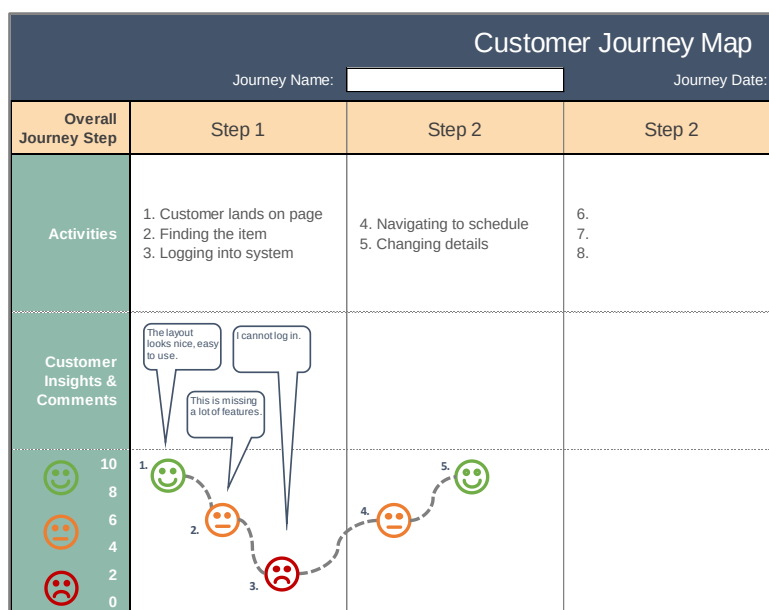


Figure 26. Customer Journey Map

27. UML Sequence Diagram

What is it?

A Unified Modelling Language (UML) Sequence Diagram shows the sequence and flow of information between objects in a system, or classes within code.

How to use it?

Start by noting all the parts that make up your system, or the multiple systems that make up the experience. If you are diagramming code, it might be the classes that make up the program. Draw your customer in the top left, then note each part in boxes from left (first) to right (last) with lines moving down from each. Starting in the top left draw an arrow from left to right (and back again where necessary) showing the sequence of information until the end of the process.

Place a rectangle over each part's line for as long as it is active – when arrows are interacting with it.

When to use it?

Sequence diagrams are used by business analysts, professionals and developers to document the requirements of a new program, or a change in an existing one.

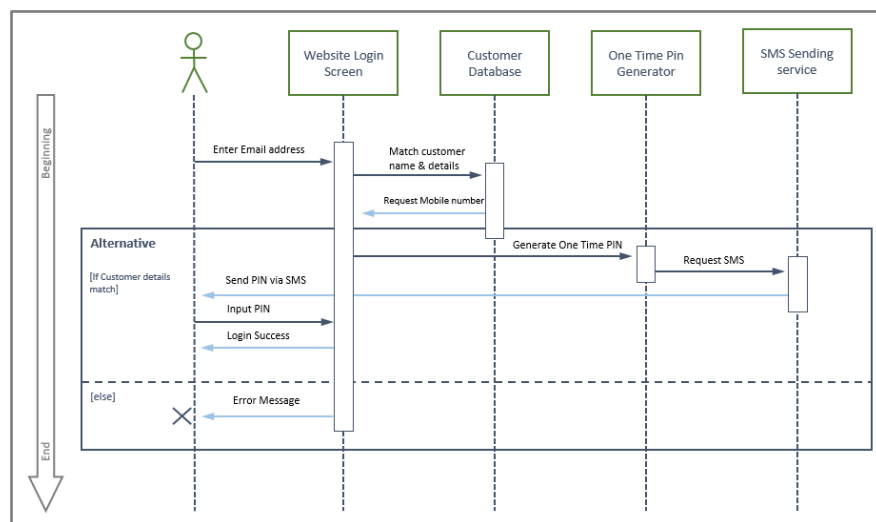


Figure 27. Customer Journey Map

Keeping a Schedule on Track

28. Schedule Network Diagram

What is it?

A schedule network diagram outlines the logical order in which different activities must be completed for your project, providing a clear sequence of activities to complete your project efficiently. It also helps us find the Critical Path, which is the shortest time our project will be able to be completed.

How to use it?

Make a list of all the activities to be completed as part of your project, and write them from left to right in order of when they need to start. Some activities can be completed at the same time (i.e. by different teams or people) so place these under each other in the same column. Other activities may not be able to be started until another activity has finished, so place these in sequence. Next, write the duration of each activity to get an idea of the length of your project. Add up the durations for each different sequence of activities – the sequence of activities that cannot be moved and that is the longest all together is the Critical Path, which is the path with no “leeway”, wriggle room or “float” to move activities back.

When to use it?

A schedule network diagram is useful when planning and then controlling the schedule of your project, as a way to visualise the activities. It allows you to see the “float” in your schedule – any activities that could be pushed back if needed. You can adjust the network diagram as the project commences based on any changes that occur.

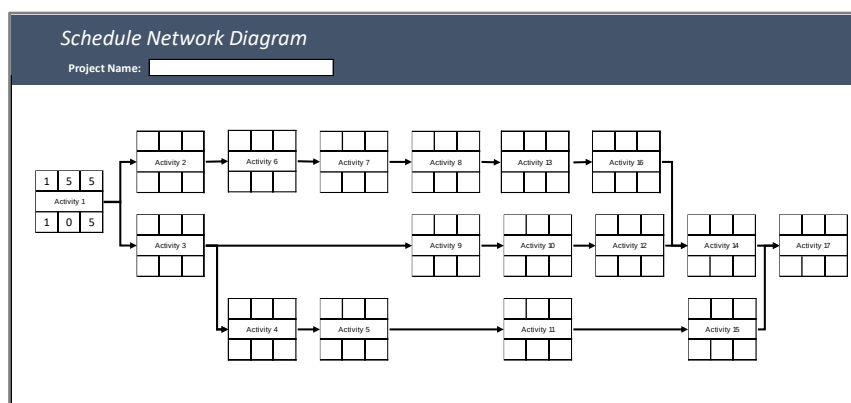


Figure 28. Schedule Network Diagram

29. Gantt chart

What is it?

A Gantt chart is a type of bar chart that shows a project schedule, first created by Henry Gantt in 1910. It also shows how project activities are dependent on each other, when one activity needs to be completed before another can begin.

How to use it?

List the project activities down the left-hand side of your page, preferably starting with the activities you know should be completed first closer to the top. List the dates (in days or weeks) up the very top. Next, draw a box from the start date to the finish date in the appropriate place for each activity. You may need to meet with different stakeholders or teams if they have input into when an activity can be completed. You can adjust your Gantt chart over time if your project schedule changes.

When to use it?

You can use a high level Gantt chart in your Project Charter when initiating your project. Once it has begun, it is useful when planning, creating and updating the Schedule over the life of your project. In many cases a change in schedule will require a Change Request to be approved by the Project Sponsor, and a Gantt chart will assist as a visual guide.

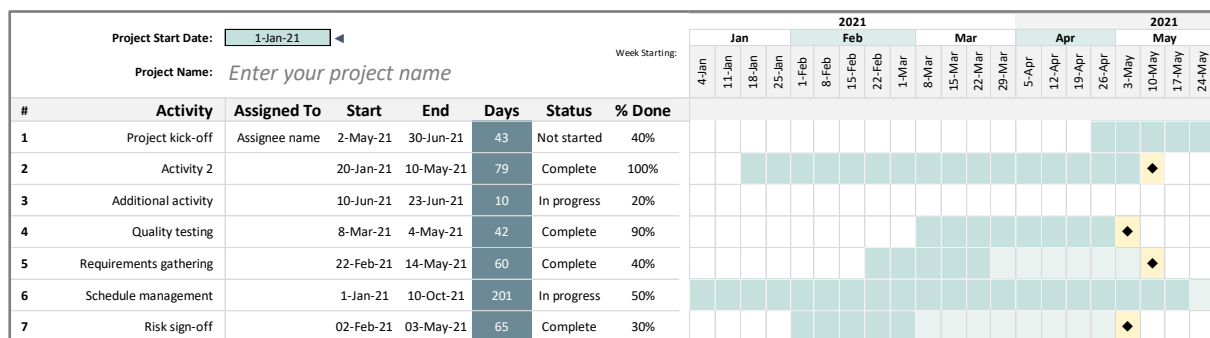


Figure 29. Gantt Chart

30. Kanban board

What is it?

A Kanban board is a tool from Lean and Agile designed to help visualize work, limit work-in-progress, and maximize efficiency (or flow). Kanban boards use cards for work tasks with the necessary information on them. The team moves those cards through various columns of their choice, such as “Backlog”, “In Progress” and “Done”.

How to use it?

Gather a list of the high level features or things you are going to deliver through your project. Then set a meeting with the teams who will be involved in the work, and for each one break it down into assignable tasks. Write these tasks on a card – one each – and place them in the first column of your Kanban board, usually named “Backlog” or “To Do”. Agree with the team and write out the remaining column names you want – “In Progress”, “Testing”, “Done” or anything else. Now the team can “pull” a card (assign it to themselves and place it in progress) when they are ready to take on new work.

When to use it?

A Kanban board is useful when managing the schedule of your project. Often used in an Agile or Lean-managed project, it also works well with a “daily stand-up”, where the team meets for 15 minutes each day and talks to the tasks they have assigned to them, takes on new work, and moves their cards across the board as necessary.

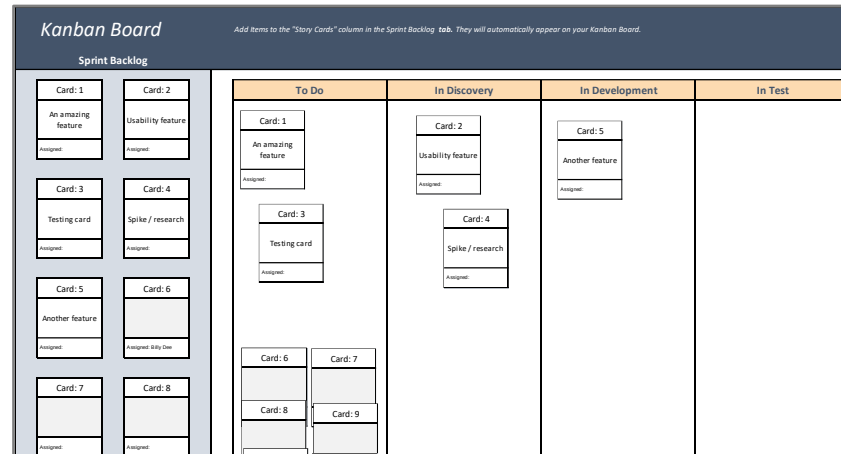


Figure 30. Kanban Board

31. Burndown chart / Burnup chart

What is it?

A burn-down chart is a technique used in Agile projects that shows a graph of work left to do (by number of tasks) versus time. The outstanding work or backlog is often on the vertical axis, with time along the horizontal. They are useful for predicting when all of the work will be completed. A burn up chart is similar, but shows tasks completed over time (adding more as more come on board).

How to use it?

Before you create your burn-down chart you will need your list of features, often approved by the Product Owner who represents the customer or who you will be delivering the value to. You will also need to have elaborated those features, and planned out the tasks (ideally of a similar size) to complete them. This will give you the total number of tasks you need to complete, which is your vertical axis. The time you wish to complete them in is up to you, but is often either one or multiple “sprints” of two to four weeks. Now create a line graph of the number of tasks remaining, against the time remaining.

When to use it?

A burn down chart is very useful when monitoring and controlling your project schedule and tasks. It is a great “information radiator” that you can keep visible at all times so everyone knows where they are up to.

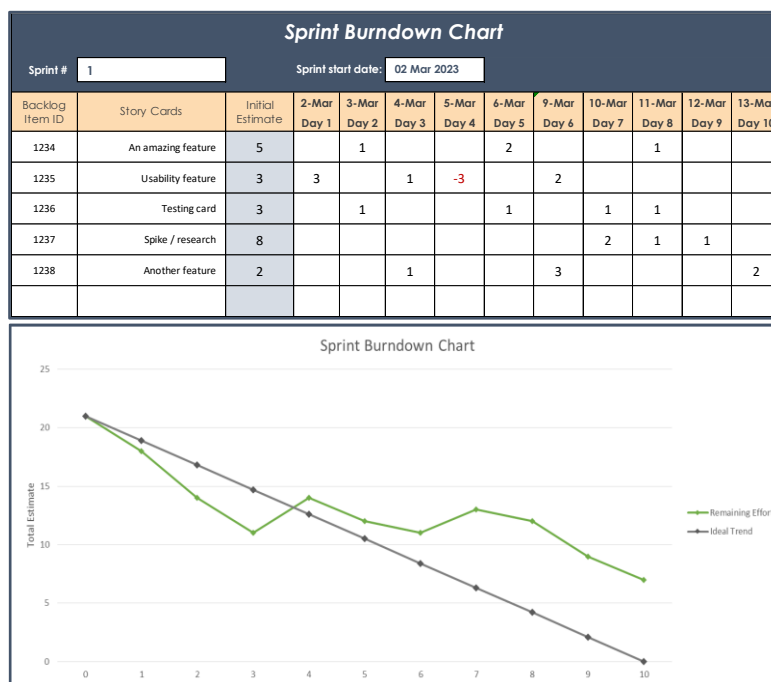


Figure 31. Burndown Chart

Managing Cost

32. Budget versus Actual Cost

What is it?

A project or organisation budget lists the costs for activities, people, systems or other resources over a period of time, often on a monthly basis. Creating a budget estimate that you will base your team or project on is a key part of management. Analysing how you are tracking against that budget in real time is also essential, to see if you are on track or need to adjust.

How to use it?

First note the area, team, company or project you wish to create a budget for. Write down a list of all the items that will cost you money, from people or labour, activities, systems, and other resources. Next, write the monthly cost for each of these items over the duration of your budget (often up to a year ahead).

Then as the year progresses, place the actual cost next to these same line items so you can see if you are on or off track.

When to use it?

Start a budget in the planning stages of your project, at the beginning of a financial year, or when kicking off a new team.

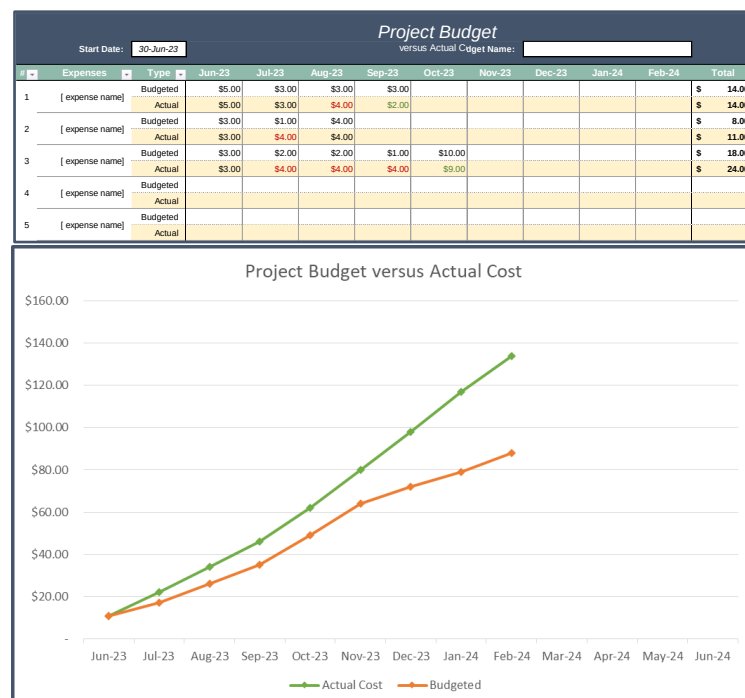


Figure 32. Budget versus Actual Cost

33. Compounding Calculator

What is it?

The power of compounding has been noted as the “eighth wonder of the world” because of its power over time. Small amounts regularly invested at a certain percentage rate of return (i.e. 10% per year from a stock market fund) can become large amounts as long as those returns are re-invested.

How to use it?

To use the compounding calculator first note your initial investment (i.e. \$1000). Next note the amount you will add each year (i.e. \$1,200 a year). Lastly note the percentage return you will likely make, based on past results (i.e. 9% per year from the stock market), and the amount of time you will invest (and reinvest) for.

The compounding calculator will show you the outcome of reinvesting your returns and gaining larger and larger returns on your reinvested capital.

When to use it?

Use the compounding calculator when you want to know how much your capital could be worth if you were to invest it (and re-invest your gains) over time.

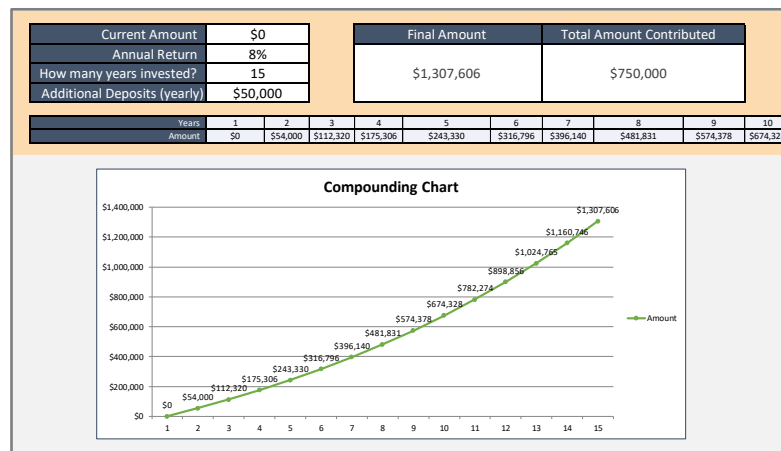


Figure 33. Compounding Calculator

Keeping Quality Under Control

34. Control Chart

What is it?

A control chart is a line chart used for showing the results and trends of a process over time. By noting upper and lower lines or control limits, it also shows when a process is within its limits or out of control.

How to use it?

First note the item you will be measuring, and the unit of measure (for example number of defects). Then create a line chart measuring those things with time on the horizontal axis moving from left to right. Calculate the average of all the measures - for example if you are using defects per day and you have 20 days, add up all the defects on your chart then divide by 20. Mark this value as the "centre line".

You can then mark the upper and lower control limit lines on your chart. You can use any value that makes sense, or is nominated by the business owner of that function. Traditionally, you can also calculate 3 Sigma away from the average line (using Excel or a calculator), but this does not fit all scenarios. A process is deemed out of control when:

- There is any point outside the control limits,
- There is a run of seven points all above or all below the central line,
- There is a run of seven points up or down.

When to use it?

A control chart is useful during quality assurance or testing of your product, when gathering defects and seeing their trend over time. It is also useful during business case and project charter creation, in analysing an existing product or process to find a business problem to solve and bring value through your project.

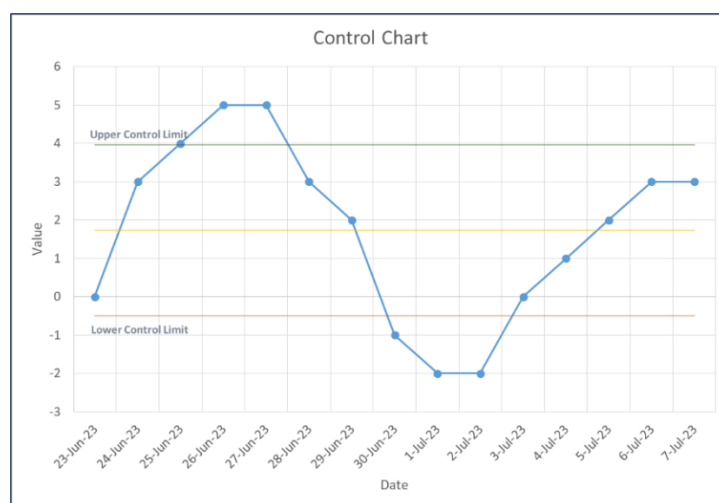


Figure 34. Control Chart

35. Check Sheets

What is it?

A checklist can also be a tally sheet, where we have certain items we want counted and a person places a mark next to them each time it occurs. This is useful in gathering data or statistics on a problem.

How to use it?

When using a checklist, first note the item or area you want to measure and the problem you are trying to solve. Write down the different situations or variables you are looking to gather data on, for example reasons behind certain defects. You may wish to have an “other” section also. It works best if either you or the person doing the work notes them immediately as they happen. Place a mark next to the item when it occurs, and tally up all the marks when you have finished. You will then be able to see the most commonly occurring ones, with the use of a Pareto Chart (also in this book) if you wish.

When to use it?

You can use a checklist when you want to measure data, making it useful during the Quality Assurance or testing phase of you project. It is also useful for showing the value of a problem when creating your business case or project charter at the beginning of you project.

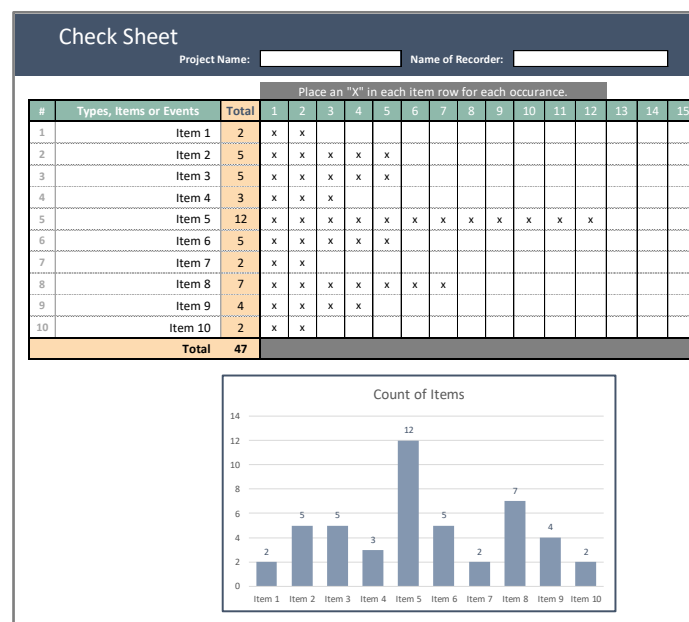


Figure 35. Check Sheets

36. Pareto Chart

What is it?

A Pareto chart is a way of visualising the most common things that occur in a set of data. It is based on the “Pareto Principle”, where 80% of the inputs were seen to create 20% of the outcomes. For example 80% of the wealth being owned by 20% of the people, or 80% of overall defects coming from 20% of the causes. It won't always be exactly 80%, but it will be around this for most things you analyse.

How to use it?

Gather the list of things you wish to measure and count (for example defects). Tally the number for each, and the total for all, and sort your areas from largest to smallest. Next, divide each item's number by the total – this will give us the percentage (for example 40 defects found by the testers, out of 100 overall = 40%). Add each percentage to the last until you get 100%. Lastly, create a bar chart for the numbers in each area, and overlay a line chart for the cumulative percentage – now you can see which items make up the majority of the outcomes you are looking for.

When to use it?

Pareto analysis is useful when analysing the most common reasons for something. For this reason you can use it when writing your business case or project charter and showing the problem your project is trying to solve. It is also useful during Risk analysis and finding the probability of something, or Quality Assurance and testing when analysing the reasons behind defects.

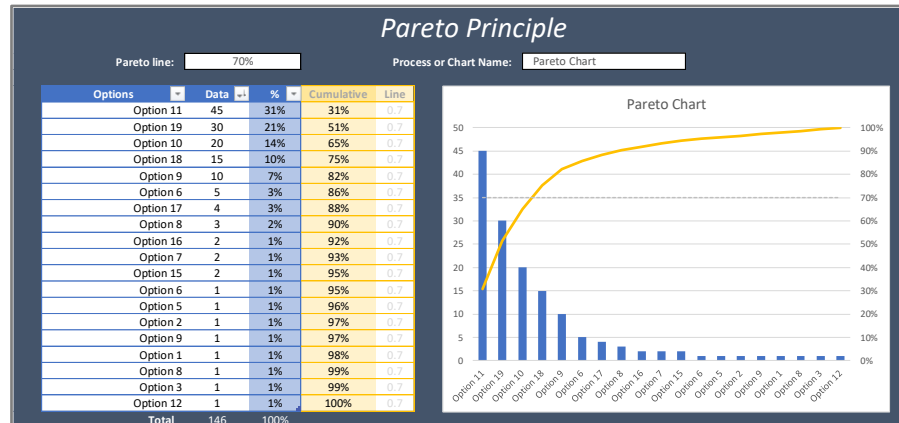


Figure 36. Pareto Chart and Analysis

37. Ishikawa Diagram

What is it?

The Ishikawa diagram, named after its creator Kaoru Ishikawa is a form of Root Cause Analysis that you will come across in Project Management. Also called a fishbone diagram or cause and effect diagram, the Ishikawa diagram articulates the problem we are looking at and then uses pre-selected categories to brainstorm reasons behind the problem. The most common categories are PIPS and MMMM:

People, Information, Process and Systems, or Man & Woman, Method, Machine, and Material.

How to use it?

Decide on the problem you want to analyse with your team, and set a meeting with the relevant people and experts from those parts of your business or project. Write the problem name in a box at the right of the page (in the “head” of the fish), then a line moving left. Draw four or more “fish-bones” off that main line, and write the categories you wish to brainstorm against, for example People, Information, Process and Systems. Now, go through the categories one by one and brainstorm reasons behind the problem, noting them on the “bone” to that category.

When to use it?

An Ishikawa diagram is useful when trying to find the root cause of a problem. Because of that, you can use it during risk and control brainstorming, during Quality assurance and triaging the reasons behind defects, or during business case creation when discovering how to solve a business problem.

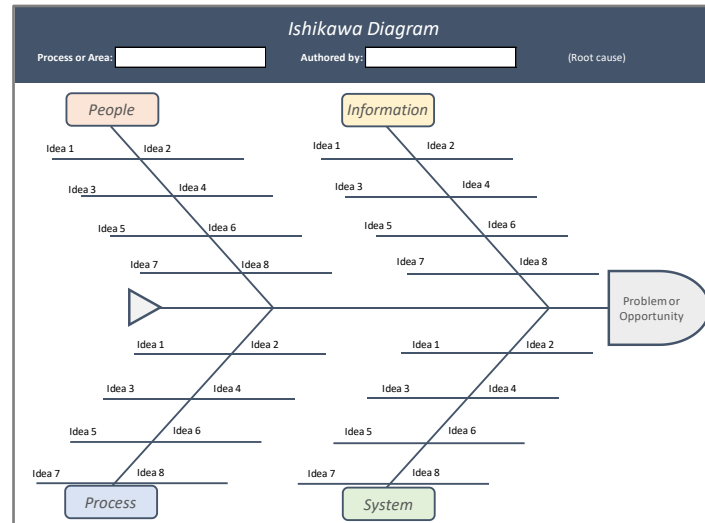


Figure 37. Ishikawa Diagram

38. 5 Whys

What is it?

The Five Whys is a method of root cause analysis taken from Lean and the Toyota Production System. It takes a business problem or defect and asks “Why?” it is occurring. With each new answer we ask “Why” again until we have exhausted our options and found the real cause. The Five Whys is useful because the first reason behind something is typically more of a symptom, and it takes a bit of digging to reveal the true problem. Remember, it may not be exactly five “Whys”, as long as you ask why enough to find the real cause of the problem.

How to use it?

Write down the problem or defect you would like to know the root cause of (for example, “Why are we receiving a large amount of defects on this particular system, or that particular building type?”). Set a meeting with the relevant experts in that area, as you will need their expert judgement. Ask your first “Why” with the group, and brainstorm reasons. It may proceed quickly as people naturally dig into the problem, so write each layer of “Why” as they occur. You may also need to go away and gather data on the problem – but ensure you regroup once you have it. When there are no more reasons, look into the last one and brainstorm ways to solve it.

When to use it?

Use the Five Whys any time you need to find the root cause of a business problem. This might be during Risk brainstorming and controls for those risks, during Quality testing and defect triage, or during Schedule or Budget management if they are going off track.

The Five Whys			
Team Name: <i>Enter your team name here</i>			
What is the problem? <i>Enter the problem or opportunity here</i>			
	Why	This happened because:	Is this the root cause?
1	Why did this happen?	Because we didn't see the item.	No
2	Why did this happen?	Because a box was in the way.	No
3	Why did this happen?	Because there is not enough room for the boxes.	No
4	Why did this happen?	Because we have too many boxes.	Yes

Figure 38. The Five Whys

Managing People and Teams

39. RACI or Resource Assignment Matrix

What is it?

A RACI is a resource assignment matrix, and it stands for who is Responsible, who is Accountable, who should be Consulted (the expert in that area), and who needs to be Informed of the outcome. It is an easy way to show who is doing what in your project (or external to your project team when engaging other areas).

How to use it?

Choose the process or mission you are assigning people for – typically this can be done for your entire project, or any other process you choose. Draw a grid or matrix, and write the tasks, actions or deliverables across the top.

Write the names of your stakeholders and team members down the left-hand side column. Now you can place an R, A, C or I in the cell that coincides with the person and the activity – this should be done with the team so everyone is in agreement or aware of their responsibilities. Remember that:

Responsible is the person who will perform or complete the activity.

Accountable is the person in charge, or who will sign off on the activity.

Consulted is the expert we can go to for their expert judgement on the activity (i.e. someone doing work in the area we are delivering to), and;

Informed is the person we need to inform when it is completed (often those affected by a change, a project sponsor, the project manager, or anyone up the project chain of command).

When to use it?

Use a RACI at the beginning of your project when you have identified the stakeholders you need to engage, and the high-level actions that need to be completed. Once you have a completed RACI, send it to the team members involved, and update it regularly as your project evolves or changes.

RACI - Resource Assignment Matrix													
Project Name:			Project Manager:										
R - Responsible - who performs the task A - Accountable - who signs off C - Consulted - who provides expert judgement I - Who is kept informed													
		Deliverables	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable
			Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable	Task or Deliverable
NAME	PROJECT ROLE												
[Enter Name or Team]	Project manager												
[Name]	Risk manager		R	C					A		C	I	
[Name]	Senior User				C								
[Name]	Quality lead			C									
[Name]	Subject Matter Expert					R							

Figure 39. RACI Matrix

40. Organisation chart

What is it?

An Organisational Chart is a hierarchical chart that uses “decomposition” – a term you will see in the PMBOK Guide (the Project Management Body of Knowledge), and it lists the people in or around your project and who they report to. It can be useful for noting the project team structure – with the Project Sponsor, Project Manager and different teams. It can also be useful for noting an existing organisational structure with functional managers and their team, to ensure you are engaging all the right people as part of your project.

How to use it?

Gather a list of people you want to turn into an organisation chart, and start with the highest ranking person. Place their name and role in a box. For your project this will usually be the Project Sponsor or the business owner. For a functional area it will usually be the highest ranking executive you will need to engage for your project.

When to use it?

Use an organisational chart when planning your project and stakeholder management. You can use a high level chart when initiating your project and creating your project charter, and an in-depth organisational chart as you plan your stakeholder management in detail, giving you clues on who you will need to engage and keep on-side during your project.

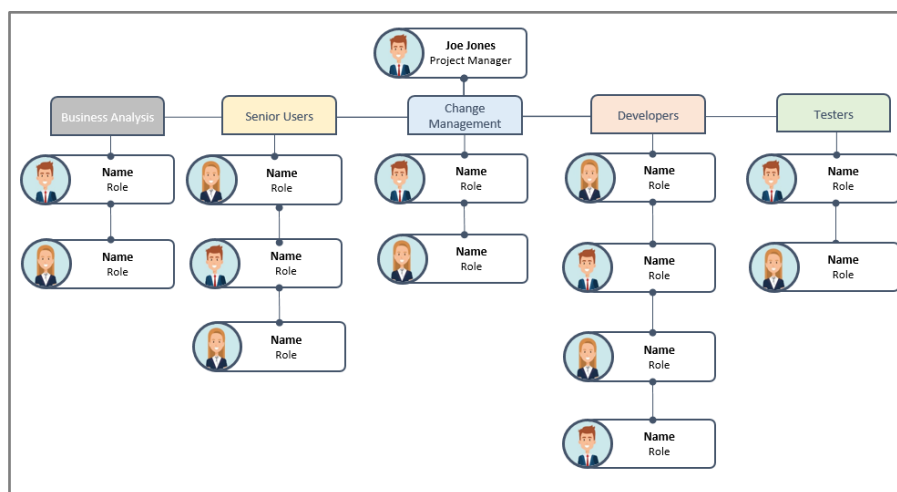


Figure 40. Organisational Chart

41. Stakeholder Assessment matrix

What is it?

A stakeholder assessment matrix is a way to visualise the people involved in your project and how engaged they are in helping you meet your goals. You can use it to note where their engagement is currently and where you would like it to be.

How to use it?

Make a list of all the stakeholders involved in your project – whether they are part of the project team or impacted by the outcome of your project. Draw a grid, noting your stakeholders down the left-hand side. Up the top note the words from left to right: Unaware, Resistant, Neutral, Supportive, and Leading. Now you can mark where each stakeholder is currently with a “C” (i.e. are they currently unaware of your project?) and where you would like them to be with a “D” (i.e. do we want them to be leading the charge, or just supportive of your project?). Typically to improve their engagement, more communication is needed in a way that suits them.

When to use it?

Use the stakeholder assessment matrix to begin planning your project, close to the beginning of your project and then monitoring it regularly as your project evolves.

Stakeholder Assessment						
Project Name:						
<i>Current state Desired state</i> <i>Place a C and D in a row for each stakeholder</i>						
Stakeholders		Unaware	Resistant	Neutral	Supportive	Leading
Team or Area 1	[Enter stakeholder here]		C		D	
	[Enter stakeholder here]			C	D	
	[Enter stakeholder here]	C				D
	[Enter stakeholder here]				C, D	

Figure 41. Stakeholder Assessment Matrix

42. Impact over Influence Chart

What is it?

An impact over influence chart is a way to visualise how much the stakeholders in our project might impact our work, and how much we will need to communicate with them. It notes the Impact on the horizontal axis – how much they are impacted by the project, and the Influence on the vertical axis – how much power they have to influence the outcome of your project.

How to use it?

Start by making a list of the stakeholders (either individuals, or teams) in your project. Next to each stakeholder, note a value of 1 to 10 for the impact your project will have on them (i.e. a small change, or a large change?) and the influence they have to help or hinder your project's goals. Then draw a graph with the title of "Impact" on the horizontal axis and "Influence" on the vertical axis, marking the values from 1 (low) to 10 (high) on each. You can now place the names of your people or teams on the graph where the two values meet.

When to use it?

Use an Impact over Influence chart during project planning, when you need to determine who to engage and how often. Stakeholders who are highly impacted and also have a large influence over the success of your project should be engaged often and in the ways that suit them.

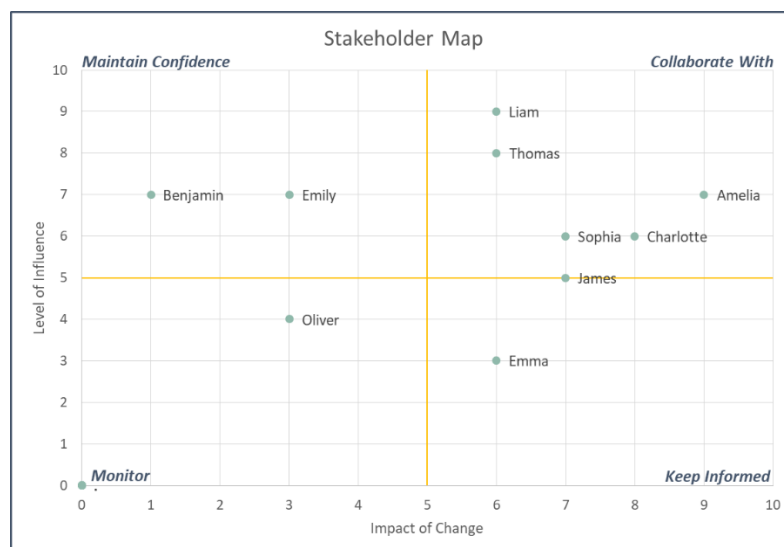


Figure 42. Impact over influence chart

43. Saliency model

What is it?

The Saliency model is a framework for analysing the importance of stakeholders in your project and how often they should be engaged. Instead of the two dimensional view of an Impact over Influence chart, it uses a three dimensional view with the following categories:

Power, where we note the power a stakeholder has to influence our project.

Legitimacy, where we note how legitimate their claim is to make changes to our project.

Urgency, where we note how often we should be communicating with this stakeholder.

Each of these categories usually has a rating out of ten, and a high rating in each category for a stakeholder helps us to know we should engage them often.

How to use it?

First gather a list of all the stakeholders that are in your project, could impact your project or will be impacted by your project. Write them down the left hand side of your page, then write “Power”, “Legitimacy”, and “Urgency” in the top row of your page. You are now able to rate each stakeholder for each item from one (low) to 10 (high), using the guide for each above. Tally the numbers together for each stakeholder and the higher the number the more you will need to engage with that stakeholder. It can also be visualised as three circles, representing each category.

When to use it?

The Saliency model of stakeholder analysis can be used right from the beginning of your project, as you gather the high level view of stakeholders for your Project Charter and initiate your project. It will be useful when analysing the Risk in your project and who needs to own the actions or controls. It should also be revisited throughout your project, especially if any changes are made to stakeholders, procured resources, schedule, cost or quality.

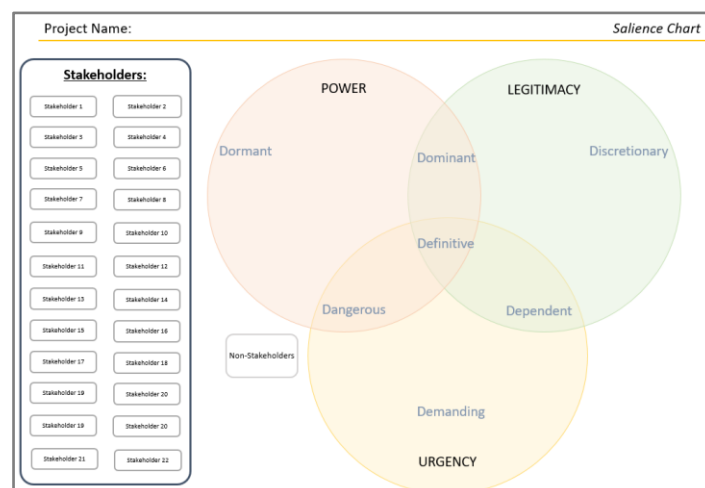


Figure 43. Saliency model

44. Retrospectives

What is it?

Retrospectives are an Agile tool for checking in on the project process and ensuring it is the best it can possibly be. It asks four main questions to gather feedback and improve:

What went well?

What could be improved?

What still puzzles me?

What did I learn?

How to use it?

At the end of a Sprint or iteration of work, usually around two to four weeks, set a meeting with your team and ask the Retrospective questions above. Note everyone's answers for each. You can use an Affinity diagram or method to group similar answers. Once you have been through each question, take the top problems (ones with the most answers grouped together) and brainstorm ways to improve them with your team. Remember to make it about the process, not the people (i.e. how can we improve our process?) and do not blame or judge answers as they arise.

When to use it?

Retrospectives can be done at all stages of your project, or when something has been completed such as an iteration, phase or milestone. Typically they are used in an Agile environment at the end of an iteration of work, but can also be used on Waterfall projects at the end of phases or the project itself as a Post Implementation Review (PIR) or Project Post Mortem.

Retrospective					
Team Name:					
Iteration:					
Successes: What went well?		Challenges: What could be improved?		Questions: What puzzles us?	
1	Enter your successes	1	Enter your challenges	1	Enter remaining questions
2	More successes	2	More challenges	2	Numbers will appear automatically
3	Even more successes				
#	Actions	Owner		Status	
1	Enter action here			Not Started	
4	Second action here			In Progress	
5	Numbers will automatically appear to the left			Complete	

Figure 44. Retrospectives

Managing Risk

45. Decision Tree analysis

What is it?

A decision tree shows us the various paths or scenarios we could take, the probability of their occurrence, any risks associated with the decision and their cost. Each decision point is represented by a box, with the various paths to new decisions stemming from each box. The end-points of branches in the decision tree represent the outcome from following that particular path, which can be negative or positive.

How to use it?

You may wish to perform this with various experts – the project sponsor or their representative, a risk expert, and other business subject matter experts. Start by brainstorming with your team and writing down all the options or scenarios available to you. For each, note the costs (or risks) associated with it and the probability of it occurring.

Your Decision Tree starts with a Rectangle representing the decision to be made. You can then branch off into various Decision nodes such as “make, or buy” (represented by a square) and Chance nodes such as “40% probability of high demand” (represented by a circle).

At the end of each “Path” place a triangle, and add up the total benefit minus the total costs to determine the outcome (either positive or negative). Finally, multiply all the Chance notes together on that path for its final probability (i.e. $0.60 \times 0.50 = 0.30$ or 30% chance of happening). We want the paths with the most positive outcome, combined with the highest chance of happening.

When to use it?

You can use a decision tree when first planning your project, during the business case or project charter and when trying to determine the best way to solve the business problem. It is also useful during Risk planning, when determining the cost of certain risks and their potential controls (paths or decisions).

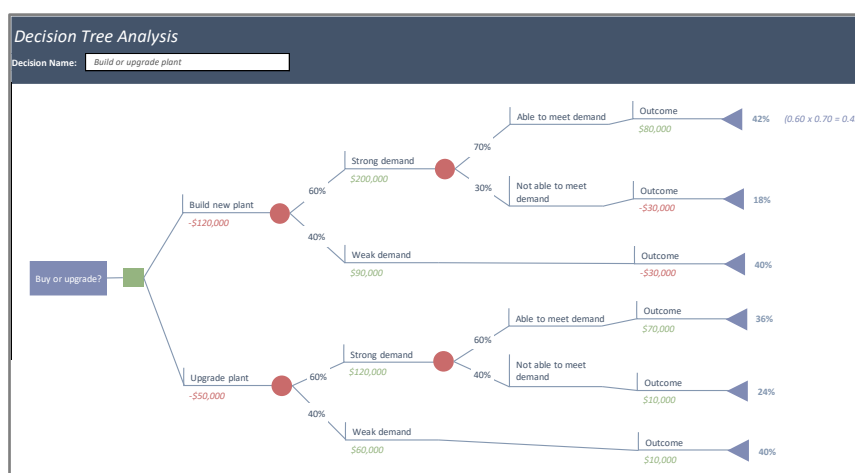


Figure 45. Decision Tree Analysis

46. SWOT Analysis

What is it?

SWOT analysis is a framework for discovering both positive and negative risks to your project. It stands for Strengths, Weaknesses, Opportunities and Threats. Risks are most commonly separated into Delivery Risks (risks to the satisfactory delivery of your project), and Delivered Risks (risks to the business or customer once your project is delivered).

How to use it?

Start by setting a meeting with your project team, including any key subject matter experts, risk experts, and your project sponsor if they wish to be involved. Note down your project's objective, and whether you are gathering either Delivery risks (risks to the delivery of your project) or Delivered Risks (risks to the organisation once the product is delivered). Brainstorm with your team the things it does well (Strengths), and things it does not do well (Weaknesses). Next note the threats you will need to mitigate and the opportunities you may wish to exploit.

When to use it?

Use SWOT analysis during project initiation to find high level risks or determine new features or scope, when creating the business case or project charter. It is also useful during the Risk knowledge area when planning and executing your project.

SWOT Analysis				
Organisation or Team:				
Internal	S	Strengths	W	Weaknesses
	1	What resources do we have available?	1	What could we improve?
	2	What do we do best?	2	What disadvantages do we have?
	3	What do our customers love most?	3	Where do we lack efficiency?
	4	What are we more efficient at?	4	Where are we wasting money?
External	5	What makes us stand out?	5	Where are we wasting time and resources?
			6	What do our competitors do better?
			7	What customer complaints have we had about our service?
	O	Opportunities	T	Threats
	1	What is missing in our market?	1	Which of our weaknesses could prevent us from meeting our goals?
	2	What could we create or do better than a competitor?	2	What obstacles do we face?
	3	What is missing in our market?	3	What changes are occurring in our market or industry?
4	What could we create or do better than a competitor?	4	Are there changes in technology that could threaten our success?	
5	What new trends are occurring?	5	What social changes or trends could threaten us?	
6	What new technology could we use?	6	Are there any policies or legislation changing that might threaten our success?	
7	What openings in the market are there?			

Figure 46. SWOT Analysis

47. PESTEL Analysis

What is it?

PESTEL analysis is a high level framework for brainstorming threats and opportunities for your project. It stands for Political, Economic, Social, Technological, Environmental and Legal, and is a great way to gather ideas of things that might impact your project or business either positively or negatively.

How to use it?

Set a meeting with your project team – the project sponsor, team leads, risk expert and anyone else you think will help. Write down the project goal so everyone is clear, then draw a grid with “Threats” and “Opportunities” written up the top, and Political, Economic, Social, Technological, Environmental and Legal down the left hand side. Now you can brainstorm both threats and opportunities for each of these and write them in the cells where they meet.

When to use it?

Use PESTEL analysis as a high level start to gathering Risks for your project, during business case or project charter creation, and throughout the early phases of your project when first analysing any Risks.

PESTLE Analysis					
Organisation or Team:					
P Political			E Environmental		
1	What trading policies impact us?		1	How can we source, trade and test our products?	
2	What regulations must we follow?		2	Is everything we do ethically sound?	
3	What will happen if another party gains power?		3	What is our carbon footprint and is it worth reducing it?	
4	What future policies do we need to be aware of?				
S Social			T Technological		
1	Who is our target market?		1	What technology is critical for our day-to-day operations?	
2	How are consumer opinions changing regarding our product or service?		2	What new technology is available that could streamline our business?	
3	Is our core demographic growing or slowing down?		3	Do we depend on 3rd parties for any tech support or solutions?	
4	How do we interact on Social Media?		4	How are we using technology to stay ahead of the competition?	
			5	What can we learn from the data we collect?	
L Legal			E Economic		
1	How are we impacted by changes to legislation and regulation?		1	What taxes impact our business?	
2	How do we keep ourselves compliant?		2	Is the economy stable, unstable, or growing for your industry?	
3	Are we up to date with data protection laws in all countries we operate in?				

Figure 47. PESTLE Analysis

48. Risk Probability and Impact matrix

What is it?

The Risk Probability and Impact matrix is a way of brainstorming and categorising project or business risks by how likely they are to occur, and the impact they might have if they do occur. For each risk that you find with your project team (or gather through a Prompt List, also in this book), give it a rating of one to five for:

The probability – how likely is it that it will occur? One is not very likely and five is absolutely certain.

The impact – what is the impact if it does occur? One is almost no impact (i.e. no cost or no disruptions) and five is a high cost or disruption.

You can also use any other scale you like, for example one to 10 or a percentage.

How to use it?

Gather the list of project risks you have with your team, whether you have brainstormed them previously or used an existing list of common risks for your organisation. Rate each one from one to five for their probability and their impact. Now multiply those two numbers together (e.g. Probability of two, and an Impact of four gives us eight). Now we can place these risks into a Probability and Impact matrix. Draw a grid with the “probability” down the left-hand side from 1 (low) to 5 (high), and the “impact” down the bottom row from 1 to 5 also. Now you can place your risks in the cell where their value coincides (e.g. where two and four meet for our example above). Risks in the top right hand corner are much higher and should be monitored carefully. You can also use names to label your risks (i.e. Sustainable, Moderate, Severe and Critical).

When to use it?

Use a probability and impact matrix when working in Risk planning for your project. You can revisit and update your matrix throughout your project as you monitor and control it, or anytime something changes for your project scope, schedule, cost or quality.

Risk Assessment										
Area Name: Project or Area			Updated by:			Date last updated: 10-Mar-23				
Project ID	Risk Description	Caused by & Consequence	Risk Owner (or Assignee Name, Role, or Sales)	Inherent Risk (before controls)			Controls (if any)	Residual Risk (with controls)		
				Probability	Impact	Risk Rating		Probability	Impact	Risk Rating
1	Risk 1	Caused by: Consequence:	Assignee Name, Role, or Sales	Very High	Medium	Severe		Low	Medium	Moderate
2	When you enter risks, the numbers to the left will appear automatically.	Caused by: Consequence:		Low	High	Severe		Low	High	Severe
		Caused by: Consequence:		Medium	Medium	Moderate		Medium	Medium	Moderate
		Caused by: Consequence:		Low	Medium	Moderate		Low	Low	Sustainable

Inherent Risk Matrix (before Controls)						
Impact		Very Low	Low	Medium	High	Very High
Probability	Very High			1		
	High	1			1	
	Medium		1	1		
	Low		1	1	1	
	Very Low					

Figure 48. Risk Probability and Impact Matrix

A Final Note

Congratulations! You have made it to the end of this workbook.

By now you have seen some of the most popular and most-used templates found in knowledge work around the world, used by leaders, project managers, analysts, consultants, improvement specialists and more.

But more than that, you have the added benefit of having these templates already created for you, ready-made and ready to use. Not only have you saved the hundreds of hours and thousands of dollars that went into creating them, but you can use them to make an unlimited amount of improvements, benefits and value in your start-up, business venture, or organisation.

On a personal note I want to say that I truly believe in you. You have already proven that you are on the path of improvement, and I believe you are destined for great things. Part of that is adding value to others through your future work, by working and brainstorming together to improve.

Best wishes,

David McLachlan

APPENDIX: Bonus Tools and Methods

Estimating Tools

What are estimating techniques?

Throughout your project you will need to estimate things such as the cost or time to complete certain items. Because of this there are various methods of estimating you can use at different times ranging from fast and high level, to longer and more detailed. In the next three sections we will look at these different methods and the best times to use them.

How to use it?

Ensure you have your list of project activities, decomposed or broken down to the team work package level (that a team can work and estimate on). Then for each activity, ask the person closest to the work (often the one who will perform it) to assist you with the estimate for either cost or time. You can then fill out the cost next to the item in your Budget, and the time on your Gantt Chart.

Analogous Estimating

What is it?

Analogous estimating is when we compare the thing we're estimating to something similar – a past project, a similar work package or task we have seen previously. It is like an "Analogy", where we compare two things that are similar, only this time we are doing it with cost and time estimations. It is easy to do, but it is also not as accurate as parametric or bottom up estimating.

When to use it?

Use analogous estimating when there is a limited amount of detailed information about the project available, for example for a quick estimation by comparing to a similar project, during the business case creation.

Parametric Estimating

What is it?

Parametric estimating is where we use a "Parameter" – a certain unit of measure against what we want to estimate. For example, \$50 per square foot, \$1000 per roll of sheet metal, or \$600 per day.

When to use it?

Parametric estimating is useful when we have more time and need a more detailed estimate, most likely during project planning and after our project is approved and initiated. We can gather parameter costs from the experts in the areas we're estimating in, or we can use public information such as databases, or we can even use past projects to gather a rough guide to the parameters we are estimating on.

Bottom up estimating

What is it?

Bottom up estimating is where we estimate on the individual work packages, tasks or activities separately – starting at the bottom of the work breakdown structure. Once each package has been estimated on, we add them together to get a final overall budget.

When to use it?

Most commonly used for estimating cost, Bottom Up estimating is the most detailed estimating method. We can also use it in conjunction with our other methods, for example the teams may estimate their work packages using Parametric Estimating (based on a unit per measure), or Analogous Estimating (based on previous projects).

Three point estimating

What is it?

Three point estimating takes an average of three points – an optimistic estimate, a most likely estimate, and a pessimistic estimate – and uses that as the estimation. Simply gather the three estimations or brainstorm them with your team, add them together then divide by three to get the average.

When to use it?

Three point estimating is useful when there are many differing opinions on what something should be. Instead of arguing we can gather the optimistic, most likely and pessimistic values and take an average.

Fist of Five (Voting Technique)

What is it?

The fist of five is a variation on voting commonly used in Agile projects. It is a way to allow a group to make a decision together.

How to use it?

The fist of five can be done in person (e.g. at a team meeting or stand-up around a Kanban board) or virtually (on a video call). Ask the question that needs to be decided, then ask each team member to vote from one to five on their level of support for it, or confidence in the idea. If a team member holds up fewer than three fingers, the team member is given the opportunity to discuss any objections with the team. The project manager continues the fist-of-five process until the team achieves consensus (everyone holds up three or more fingers) or agrees to move on to the next decision.

When to use it?

You can use the fist of five any time a decision needs to be made that includes a level of support and discussion around the idea. Deciding what feature to work on next, the likelihood of a risk occurring, or even the project methodology can all be done using the fist of five.

Communicating and Negotiating

The Assertive Inquiry

What is it?

The Assertive Inquiry is a technique used by Allen Lafley, the former CEO of Proctor & Gamble. In the book "Playing to Win" he describes using this technique to change the culture of his senior leadership team and move the company forward. It involves asserting your position on something "This is how I see it," and then enquiring if others see it the same way, "Is this your experience too?" By doing this you are inviting feedback when it may not have been forthcoming normally.

How to use it?

When you are running a meeting or speaking with your team or stakeholders, and you have a point of view you need to get across to everyone, state your point of view clearly then ask for feedback with phrases such as:

"Is that how you see it too?"

"What is it in your experience?"

"Am I missing anything here?"

"Can you see any ways this could be improved?"

Listen carefully to the answers people give and take them on board.

When to use it?

Use the assertive inquiry when you are running a meeting, speaking with stakeholders, gathering ideas or requirements or scope, risks or costs or any other time throughout your project. Project managers spend more than 80% of their time communicating, and this is an effective way to communicate!

The Five C's

What is it?

The Five Cs are a way to ensure your communication meets the needs of your intended audience and has the desired effect. It is particularly useful in written communication but can work just as well when presenting a speech, running a meeting or making an introduction. The Five Cs are:

- Clear purpose and expression directed to the needs of the reader,
- Concise expression (elimination of excess words),
- Correct grammar and spelling,
- Coherent and logical flow of ideas,
- Controlling the previous four to deliver your message.

How to use it?

If you are writing an important email or putting together your project management plan (or subsequent plan i.e. Quality Management Plan) for review, go through the Five Cs. Start with a clear purpose – what outcome do you want from this communication? This about how you can get your message across in the fewest sentences possible – the more someone has to read, the more they will drift off and miss things. Ensure your message flows naturally, with any necessary information leading up to the main topic of your message. Finally, ensure you run a spelling and grammar check – often automated through the software you are using.

When to use it?

Use the Five Cs anytime you are writing a communication to your project team, your project sponsor, the customer or business teams receiving your product, or in any document relating to your project such as the Project Charter or Project Management Plan.

Running Effective Meetings

What is it?

You will see “meetings” listed as a tool and technique in almost every knowledge area in the Project Management Body of Knowledge – it is the third most common thing after the need for Expert Judgement and Data Analysis. A meeting is getting the appropriate people in a room (physically or virtually) to gather their input or make a decision. An effective meeting is the difference between a well-run project staying on track or a potential disaster.

How to use it?

There are certain steps to a well-run meeting:

When setting the meeting, have the outcome or objective you want clearly listed in the meeting invitation. If needed, elaborate on this with an agenda of things you will be going over during the meeting. For every point during the meeting, ensure you ask for everyone's input before moving on – it is common for some people to remain quiet unless asked directly, while others will make their voices heard by being overly assertive. Ensure you stick to the allotted time by periodically reminding the group how long you have to go (i.e. “I'm conscious we have five minutes left, so let's work on our last item.”) then summarise the decisions and actions you have assigned to the group at the end. Send out these decisions and actions to the group in writing (often via email) and ask if there is anything you have missed to ensure agreement and it is correct.

When to use it?

Meetings are used throughout your project in almost every aspect – initiating your project, gathering requirements and scope, deciding the quality strategy, brainstorming risks and controls, meeting with third parties, gathering sign-off on deliverables and much more.

Negotiation – FBI Negotiating Method

What is it?

Negotiation appears everywhere throughout your career. One of the best negotiating books out there is from a former FBI Negotiator who actually created his own method. Here are five key actions you can take, from “Never Split the Difference” by Chris Voss:

1. Genuinely apologising and asking: “I’m sorry, but how am I supposed to do that?” won Chris more negotiations than any other phrase.
2. If conversation becomes spirited, label their emotion, and then pause and let them speak: “It sounds as though you are angry.” In doing this they will reveal what they truly want.
3. Mirror the last three words to get them talking even more – “I want to have that getaway car.” “That getaway car?”
4. Be oddly specific with what you ask for - \$5,326 is more believable that you have done your research, than \$5,000.
5. The real words you’re looking for aren’t “*You’re* right”, they’re “*That’s* right”. When someone says “That’s right” you have hit the nail on the head.

How to use it?

It is good to be aware of any negotiating techniques in all situations. While it’s not healthy to think that everyone is trying to get one up on you all the time, being educated about these things does put you one step ahead in the event someone *is* trying to get one up on you, and allows you to figure out win-win situations for your team as well.

When to use it?

Use negotiating throughout your project and career, such as when you need resources from certain areas, when you are procuring items from third parties, when you are going through the quality or defects and even when you need your team to do something. Saying the right things and speaking in the right way can save you time and money when done well.

Negotiation with BATNA

What is it?

Roger Fisher brought the BATNA into the world with his book “Getting to Yes”, and it stands for the “Best Alternative To a Negotiated Agreement”. Similar to the information you will get out of the “Never Split the Difference” methods, a BATNA is any other thing that the other person wants, that isn’t directly related to price (or the actual thing you’re negotiating for). For example if someone in a functional area does not want to release resources, and you elicit more information using Chris Voss’ techniques, you might find that they actually want more of the glory for themselves, or another project has taken their best person, or they don’t actually understand the change. All of these can be solved without needing to get into a fight about the resource.

How to use it?

Use the methods from “Split the Difference” (also outlined in this book) to elicit more information from the people you need something from. Asking open questions (questions that don’t result in just a yes, no or one word answer, such as “What steps do we need to take to move this forward?”) or using Chris Voss’ techniques can help you find the things they care about that you can use as an alternative to money or the negotiated item.

When to use it?

Use negotiation and the BATNA any time you are interacting with your team, asking for resources, working with the third party to deliver something, or trying to get someone outside of your project to assist. Understanding what someone is really after is never a bad thing, and neither is getting to know people well enough to have that information. In the end, communication and stakeholder management make the biggest difference to your career success.

The Six Weapons of Influence

What is it?

The six weapons of influence were introduced by Roberto Cialdini in the 1980s, where he discovered six key ways that people gain influence in certain situations. They are:

1. Liking, where people are more likely to do something for somebody they like, or who is likeable.
2. Reciprocity, where people feel obliged to do something for you if you have done something for them.
3. Commitment and Consistency, where people are more likely to follow through on a larger promise, after they have committed to a smaller promise.
4. Authority, where people are more likely to follow the lead of someone with authority (the right job title, qualifications, or even just wearing a suit and looking the part).
5. Social proof, where if people see other people doing it, they are more likely to do it too.
6. Scarcity, where people are more likely to want something if they can't have it or it is rare.

How to use it?

Keep each of these in mind when you are working with your stakeholders or even third parties to gather resources for your project. You can also brainstorm ways you might want to use them, with your project team.

When to use it?

The six weapons of influence are useful at all stages of your project, from project initiation through to scope validation and closing a project or phase, but especially when dealing or communicating with people and needing to gather resources from areas you don't own or control.

Organisational & Management Theories

Maslow's Hierarchy of needs

What is it?

Maslow's Hierarchy of Needs is an organisational management theory you will find useful when managing the resources in your team. It comes from Abraham Maslow's Theory of Human Motivation, and states that people meet certain basic needs before they can move up to higher order needs (like achievement or belonging). In theory, a person will not move up to a higher need until the previous need is met in some way. The order is:

1. Physiological needs, such as food, water and rest.
2. Safety needs, such as feeling secure and personal safety.
3. Belongingness, such as love, friendship and belonging to a team or group.
4. Esteem needs, such as achievement and accomplishment.
5. Self-actualisation, such as a higher purpose or meaning for your life.

How to use it?

Remember when you are managing or coordinating a team that they will often need their safety needs met (assuming they already have food and water and rest) before they can move up to higher order needs such as belonging to your team and achieving within your team. This means it is important to give clarity to everyone with clear roles and responsibilities, and to protect them from any external volatility (changing company strategies, volatile stakeholders) that may affect that safety. It also means psychological safety, where they feel as though they can raise issues without reprisal. A leader can often show their own vulnerabilities or imperfections, while still focusing on solving problems and getting the work done, to help others feel better about their own mistakes or faults.

When to use it?

Use Maslow's Hierarchy of Needs any time you are managing, or a part of, a team. Remember, you do not need the title of "leader" in order to lead, and by helping your team meet their hierarchy of needs you are demonstrating leadership regardless of title.

Herzberg's Theory of Motivation

What is it?

Herzberg's theory of motivation is an organisational management theory that focuses on two different ways that people are motivated, what he called "Hygiene Factors" and "Motivational Factors". Hygiene factors will leave your team dissatisfied if they are not there, but will not lead to additional motivation. Motivational factors will lead to greater motivation, but only once the Hygiene Factors are met. They are:

1. Hygiene factors: job security, salary (especially compared to others), work conditions, vacations, status within a company.
2. Motivational factors: recognition, achievement, personal growth, responsibility, opportunity to do something meaningful, involvement in decision making, sense of importance to an organization.

How to use it?

When you are leading a team, ensure you are aware and have checked off the hygiene factors for your team's involvement. Do they have job security? Is their salary enough, or at least comparable to their team mates? Are their work conditions acceptable with enough light, water, and can they work in comfort? Then you can focus on building the motivational factors to ensure you get the most out of your team, such as giving recognition frequently, focusing on your team's achievements, ensuring your team members have a clear path of growth (working towards their goals or promotion) while working in your project, and tying the project goals to a higher meaning such as the positive impact on customers.

When to use it?

Use Herzberg's Theory of Motivation any time you are managing people, remembering that you do not need to have the title of "leader" in order to lead, especially when you know these organisational management theories that can help your team regardless of your role.

Macgregor's Theory X and Y

What is it?

Theory X and Theory Y were created by Douglas MacGregor as an organisational management theory that states there are two main different management styles. Theory X is the more authoritarian style of management, including micromanaging and “hands on” to ensure the work is done properly. Here the manager believes their team is lazy, will avoid work if they can and has little motivation. Theory Y on the other hand is an inclusive and participative style of management, where you believe that your people take pride in their work and see it as a challenge, and you trust the team to take ownership of their work and do it effectively by themselves.

How to use it?

While the participative Theory Y might seem like the ideal method to use, in reality you may need both Theory X and Y in your work. Some team members may not pull their weight, may not enjoy the work and may need to be supervised at every step or coerced into doing the job. Even if you start there, you can still move towards the other side, where those same people may also show Theory Y tendencies if they're working on something they are good at or enjoy, and be more involved in decision making, take ownership for their work, and solve problems creatively.

When to use it?

It is good to be aware of MacGregor's theories of X and Y when you are leading a team or co-ordinating people, even if it is only to know that there are always different types of people involved with different wants or needs and things going on in their life.

Parkinson's Law

What is it?

Parkinson's Law states that work will expand to meet the time (or budget) allocated to it. In other words, if you give a task to someone to complete in two weeks, and they know they can complete it in one week, it will still most likely take two weeks as they believe they have time. The work somehow becomes more complex, other things pop up or interrupt them, and extra scope and activities come into play.

How to use it?

Being aware of Parkinson's Law may not be enough to keep it at bay during your project. There are a few things you can do to avoid it impacting you, and many of them revolve around Agile management. You can focus on releasing complete, usable features in short iterations of two weeks, or release a "minimum viable product" to market to gather feedback without adding all the bells and whistles. You can check in every day with a daily stand-up, and make the work visible with a Kanban board. And never, ever set long timeframes for tasks.

When to use it?

Be aware of Parkinson's Law throughout all stages of your project, from initiation to project closure, and especially during execution of the project work and managing and leading your project resources.

The Peter Principle

What is it?

The Peter Principle states that in a hierarchical organisation, people tend to rise to their level of incompetence. The idea behind it is that a person who is good at their job will receive a promotion to a job that requires different skills (such as leadership). If that person is competent in that new role, they will be promoted again, and so on until they are promoted to a job they are not competent in, and reach their limit.

How to use it?

Simply being aware of the Peter Principle when you interact with others and lead your project teams is useful. Helping your team mates with a standard process, training, and organisational process assets such as templates will help people remain at a high level, even when they are out of their depth.

When to use it?

Being aware of the Peter Principle is handy when you are managing and leading project teams and resources, hiring or procuring resources, or promoting within your project structure.